PRODUCER GUIDE

Ratio Viña 0.3.0 · Ratio Complexus

Your vineyard's knowledge, *from cutting* to the grapevine atlas

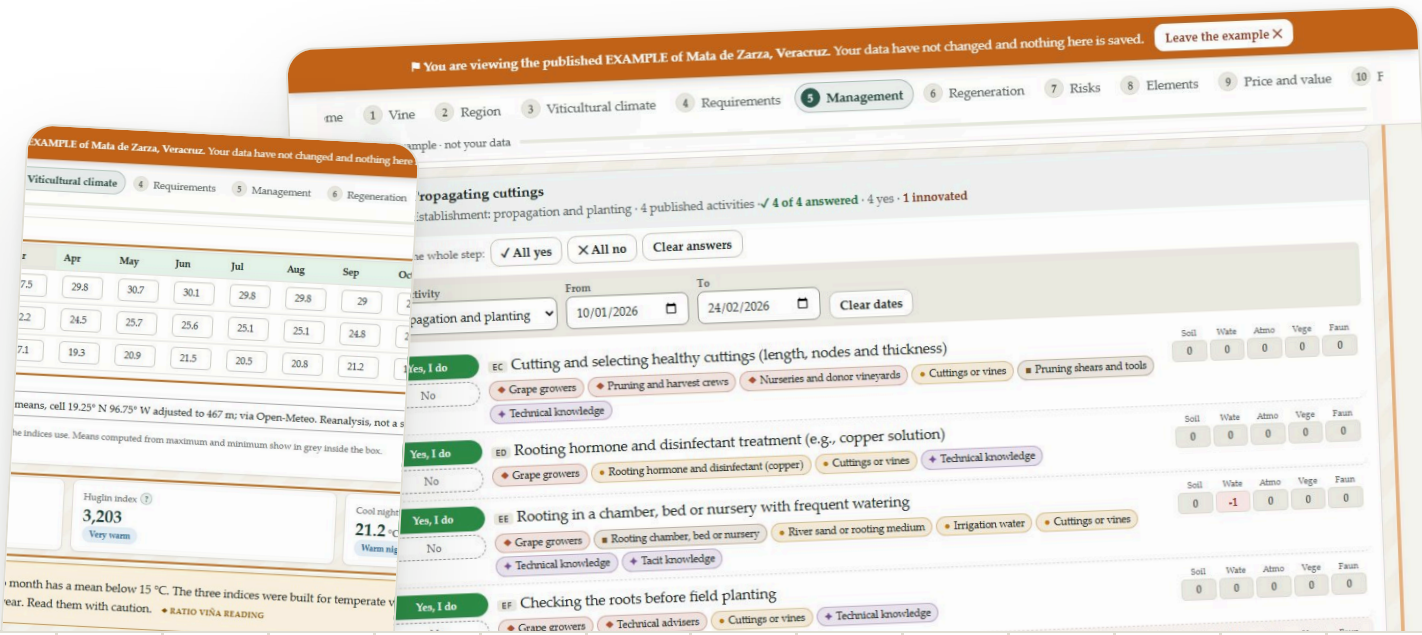
A practical guide to documenting a vineyard's local knowledge with Ratio Viña —how it was established, which variety and climate it has, how it is managed, what regenerates it, what puts it at risk and where it is heading— and to combining many vineyards into a national atlas. With a real case as a worked example: establishing table grapes in Mata de Zarza, Veracruz.

1st edition · October 2026 · covers Ratio Viña 0.3.0

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There is a companion guide for research and teaching: the Scientist Guide.



1 Vine · 2 Region · 3 Viticultural climate · 4 Requirements · 5 Management · 6 Regeneration · 7 Risks · 8 Elements · 9 Price and value · 10 Formulation · 11 Dashboard · 12 Sustainability · 13 Outlook · 14 Report · 15 Atlas

BEFORE READING

How to use this guide

Who it is for, how it is organised and what each sign on the page means.

This guide is for whoever knows a vineyard, or wants to start one, and wants to see it whole: growers, field technicians, extension agents, students doing their social service or thesis in a community. You need no coding or statistics; you need to know your vineyard and the people who work it.

It follows the app's order: one chapter for each block in the top bar, 1 to 15, each with its own colour. Every example uses **a single real case**: a **1 ha cooperating plot in Mata de Zarza, Puente Nacional, Veracruz**, where local farmers and Chapingo Autonomous University introduced table grapes in January 2025. The case is published in the *Practical guide to establishing grapes in Mata de Zarza, Veracruz* (Mojica-Zárate et al., 2026). Veracruz has no commercial vineyards yet, so this case documents something new: how a vineyard is established in a warm climate, from cutting to first pruning.

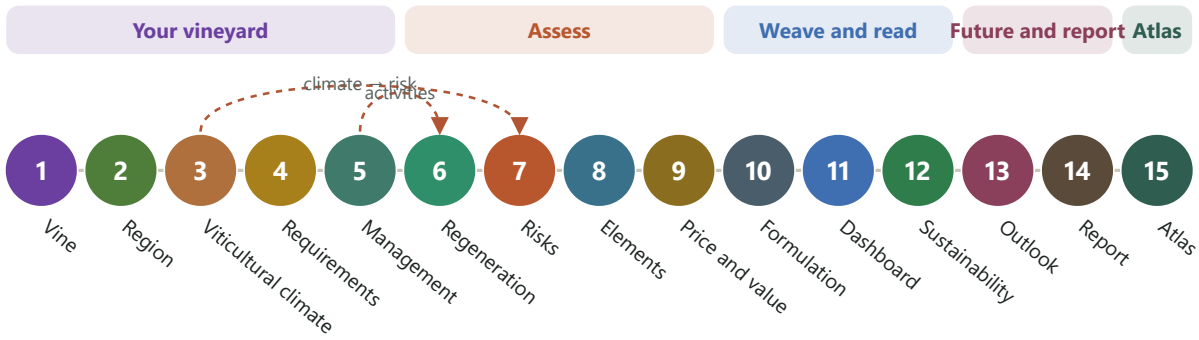


FIGURE P.1 **The fifteen blocks.** Blocks 1–5 describe your vineyard; 6–9 assess it; 10 weaves the network, which you read in 11 and 12; 13 looks to the future, 14 sums everything up and 15 combines many vineyards into the national atlas. Dashed arrows are examples of one block feeding another.

Conventions

WRITTEN AS	MEANS
Weave the system	An app button or field, with its text exactly as on screen.
Report › Save project	A path: open the first item, then the next.
2381	A value you type as shown.
1	A marker on a screenshot; it is explained below with the same number.
• Dato publicado · ◆ Lectura de Ratio Viña	The app's own labels: a figure taken from a cited source, or an app proposal you can change.

The five boxes

◆ **IN MATA DE ZARZA**

The worked case, step by step, with the real numbers the app returns.

✓ **TIP**

Shortcuts and good practice that save time or avoid mistakes.

! **CAUTION**

Common mistakes that can lead to a wrong conclusion.

? **WHY?**

The reason behind a recommendation, in a few lines. You can skip it.

→

DECISION RULE

A short table: if you see this, do that.

?

RATIO VIÑA DOES NOT MAKE UP DATA

Every figure in the app carries its source. What the app proposes is labelled ◆ **Ratio Viña reading**, and what is not documented shows as **no data**. This guide follows the same rule. The case data come from the published Mata de Zarza guide; three answers (mountain microorganisms, weeding with a brush cutter and pruning counted as winter pruning) are a personal communication from that guide's author; temperatures come from the ERA5 reanalysis, because there is no weather station in the locality; and the risk values are a Ratio Viña reading of the guide's text. Whenever something is not a published figure, we say so.

GETTING STARTED

0 Open the app and meet the screen

Ratio Viña is a web page that runs on its own in your browser. Nothing to install, no account, and your data goes nowhere.

0.1 Opening Ratio Viña

Go to **hectortecumshe-ai.github.io/RatioVina** from a computer, tablet or phone, or scan the QR code on the "Ratio Viña on one page" page. It downloads a single file; after that it works offline while the tab stays open, and you can save the page (Ctrl + S) to use it in the vineyard with no signal.

? **WHERE IS MY DATA KEPT?**

Only in the browser you use. Nothing is uploaded to the internet or to GitHub. To move it to another computer or share it with your adviser use Report › Save project (a `.vina.json` file) and on the other computer Report › Open project.

0.2 The home screen



FIGURE 0.1 **Home.** Everything you need to get around is in the top bar.

1. **ES / EN** switches the language of the whole app, report included.
2. **Producer / Scientist** switches the view; your data is kept.
3. ☀ / ☾ switches between day and night themes.
4. **?** opens the general help. Every figure also has its own **?**.
5. The **block bar**, 1 to 15, plus Validation and Theory. You can jump to any of them; next to each block you fill in, a mark says whether it is pending (○), in progress (◐) or complete (●).
6. The **pill** reminds you which vineyard you are weaving; in a new project it says "New vineyard · no data yet".

0.3 The "?" helps

Next to every important figure there is a **?**. Tap it to see what it means, how it is obtained and how to read it, with a traffic light: green (good), amber (check) and red (act).



FIGURE 0.2 **Help for a figure.** The "?" of Winkler degree-days, in Block 3.

0.4 Producer and scientist views

The **producer view** shows the essentials in plain language; it is the one this guide describes. The **scientist view** adds full tables, formulas, correlations and validation. You can switch at any time.

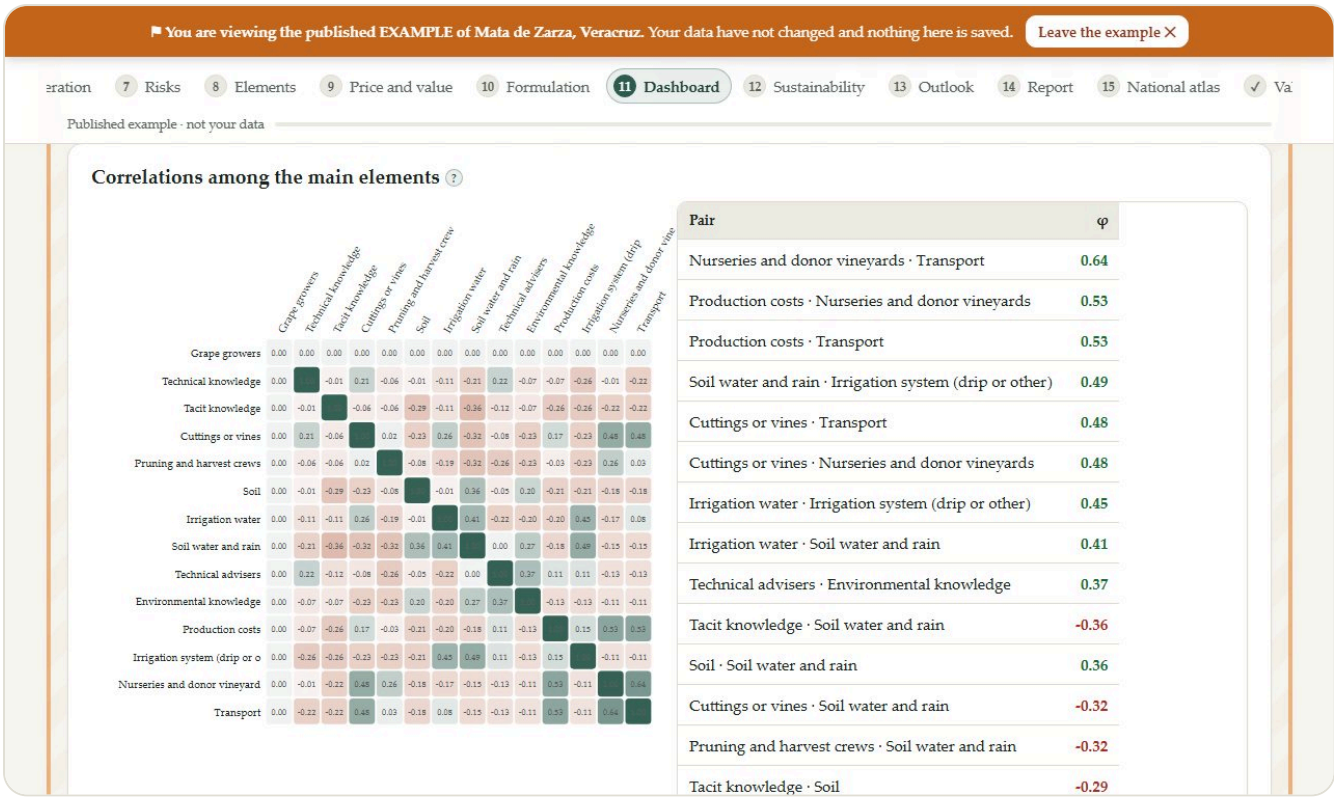


FIGURE 0.3 **Scientist view.** A Dashboard table shown only in this view.

0.5 The full route

BLOCK 1 · 5 min
Vine
Your vineyard and its history

BLOCK 2 · 2 min
Region
State, municipality, locality

BLOCK 3 · 5 min
Viticultural climate
Winkler, Huglin and nights

BLOCK 4 · 3 min
Requirements
What the vine needs

BLOCK 5 · 12 min
Management
From cutting to harvest

BLOCK 6 · 4 min
Regeneration
Regenerating or wearing out?

BLOCK 7 · 6 min
Risks
Likelihood × impact

BLOCK 8 · 4 min
Elements
Who and what takes part

BLOCK 9 · 6 min
Price and value
What it costs and what it is worth

BLOCK 10 · 1 min
Formulation
Weave the system

BLOCK 11 · 5 min
Dashboard
Network, cladogram, calendar

BLOCK 12 · 4 min
Sustainability
Soil, water, SDGs, footprint

BLOCK 13 · 5 min
Outlook
At 2, 5 and 10 years

BLOCK 14 · 2 min
Report
PDF, CSV and project

BLOCK 15 · 5 min
Atlas
Many vineyards, one atlas

✓ **IF THIS IS YOUR FIRST TIME**

The app starts blank. To first see a complete vineyard, tap **See the published case (Mata de Zarza)** on Home: you enter **example mode**, walk through the blocks with this guide and leave with **Leave the example**. Then you fill in yours from Block 1.

0.6 Yours, suggested and the example

So that you never mistake an example for your own data, the app tells three things apart at a glance:

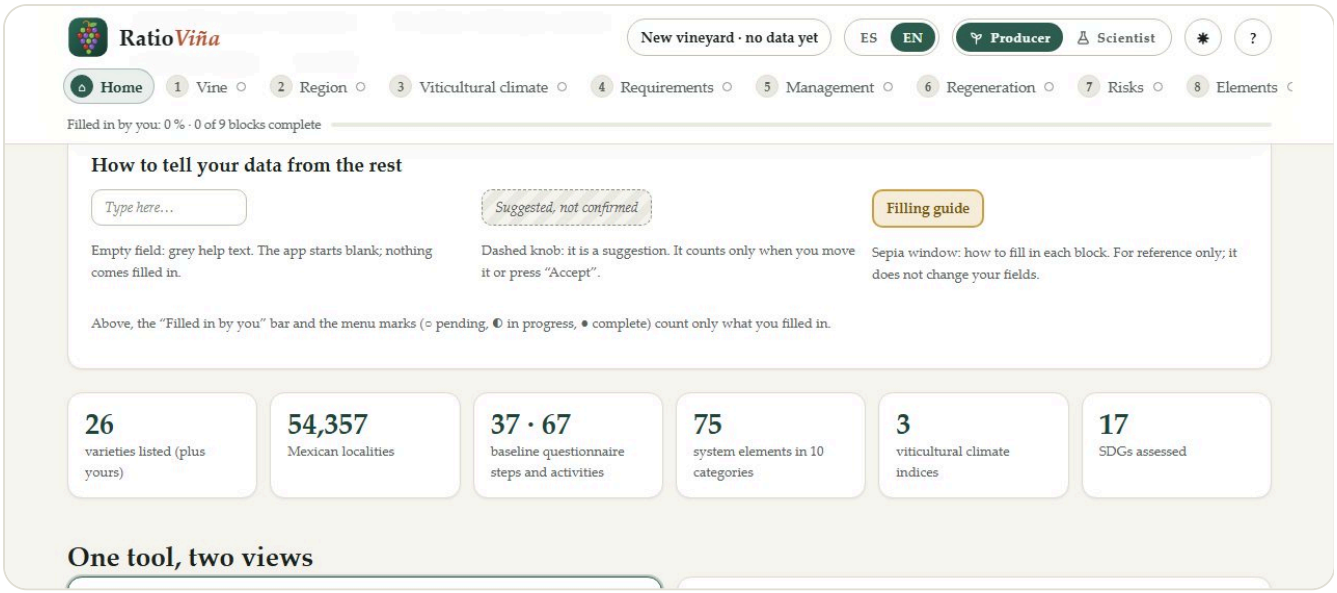


FIGURE 0.4 **The guide on Home.** Empty field, suggested value and published example.

WHAT YOU SEE	WHAT IT MEANS
Grey italic text inside a field	The field is empty; it is only a hint of what to write. Nothing comes filled in.
Dashed grey knob, "Suggested, not confirmed"	The app proposes a value from your data. It counts as yours only when you move it or tap **Accept suggestion**.
"Filling guide" window with the "GUIDE · REFERENCE ONLY" watermark	What to enter in the block, where to find the data and how to fill it in. When you close it, your fields stay the same.
Orange band at the top of the screen	You are in example mode, viewing the Mata de Zarza case. Nothing is saved; **Leave the example** takes you back to your project.



FIGURE 0.5 **Filled in by you.** The bar and the menu marks count only what you filled in.

1. Marks per block: pending, in progress or complete.
2. Total share filled in by you.

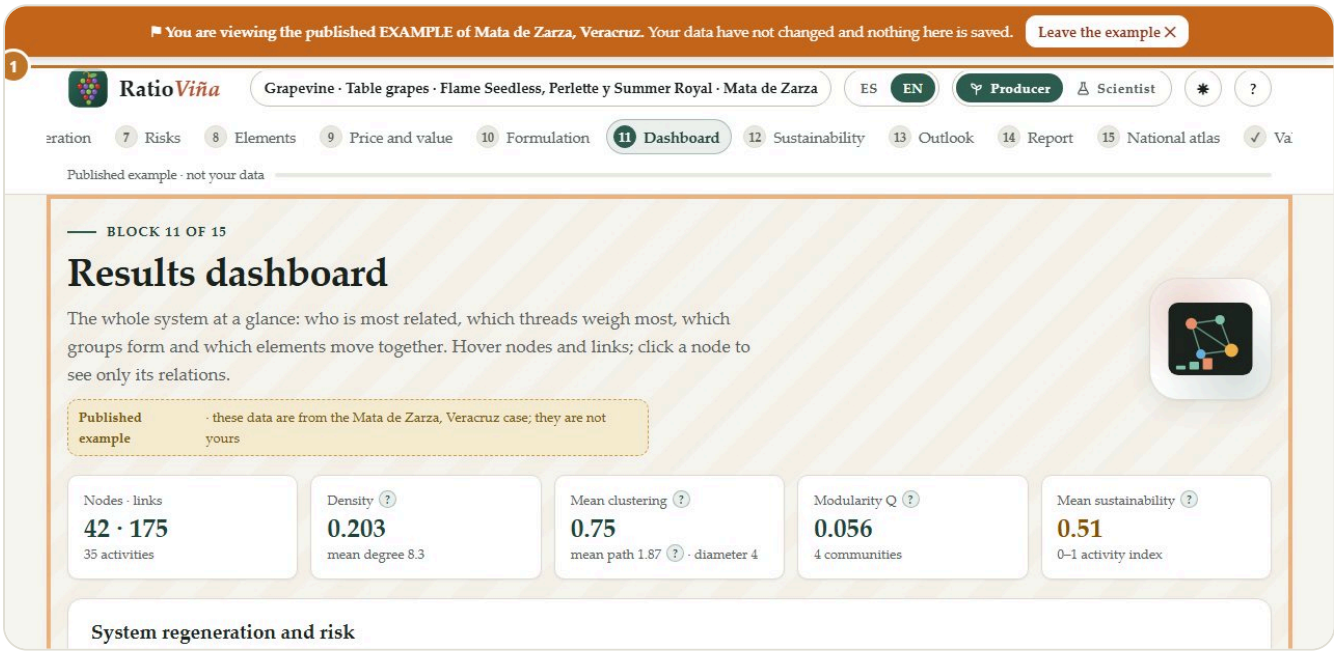


FIGURE 0.6 **Example mode.** The orange band reminds you that those data are Mata de Zarza's, not yours.

> **?** **WHAT IF I OPEN A RESULTS BLOCK WITH NO DATA?**
>
> Formulation, Dashboard, Sustainability, Outlook and Report do not show empty results: they say which blocks are still to be filled in and take you there.

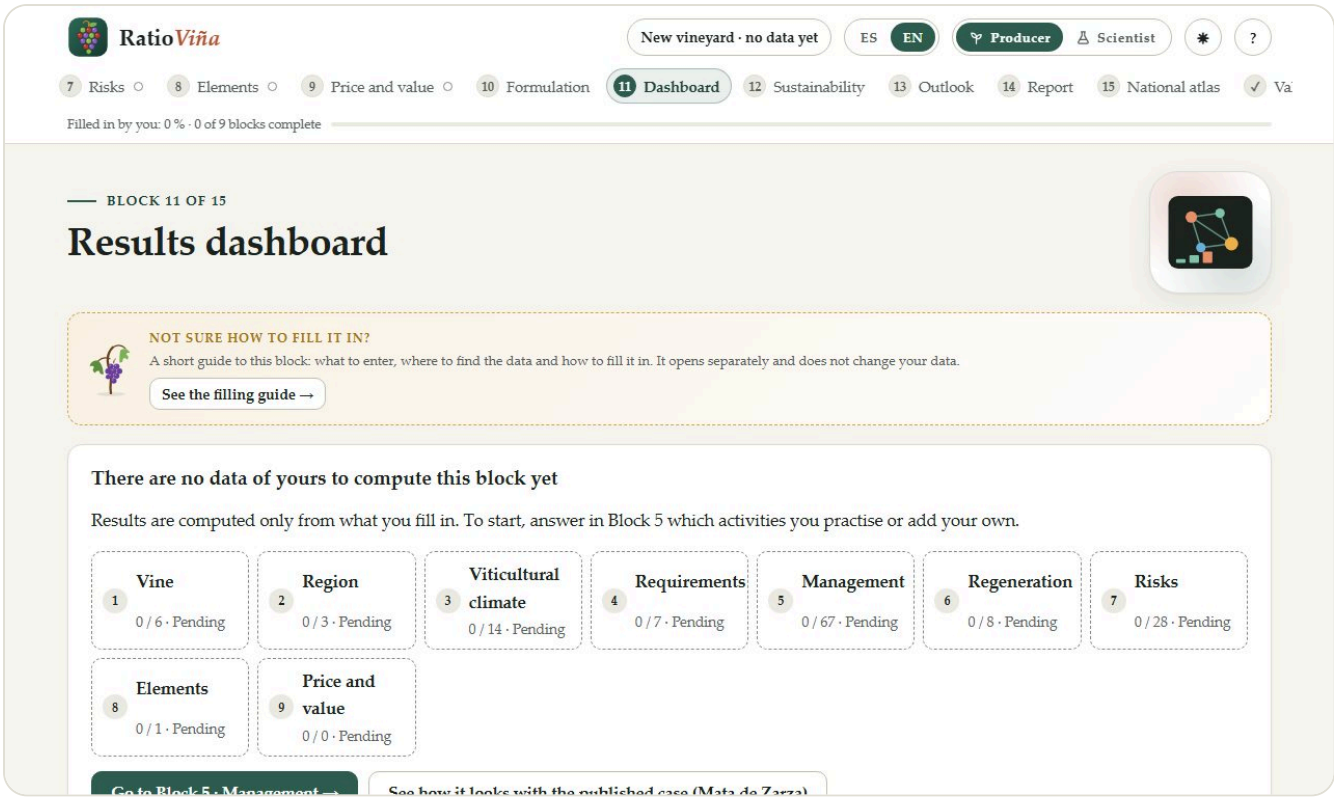


FIGURE 0.7 **A results block with no data.**

BLOCK 1 · VINE

1 Your vineyard and its history

Ratio Viña works with a single crop, the grapevine. Here you say what you grow grapes for, which variety and rootstock you use, how you train the vine, when and with what it was planted and when it has been pruned.

1.1 Use, variety, rootstock and training

First choose your grapes' **use**: table, wine, raisins, juice or several. That use shapes some Block 5 questions; for example, girdling and packing belong to table grapes and regulated deficit irrigation to wine grapes. Then type or tap your **variety** and **rootstock**; if the vine is on its own roots, choose "Own-rooted (no rootstock)". The lists are indicative: check the name with your nursery or association. Finally choose the **training system** (vertical trellis, cordon, pergola, lyre, bush vine or other) and enter vines per hectare.

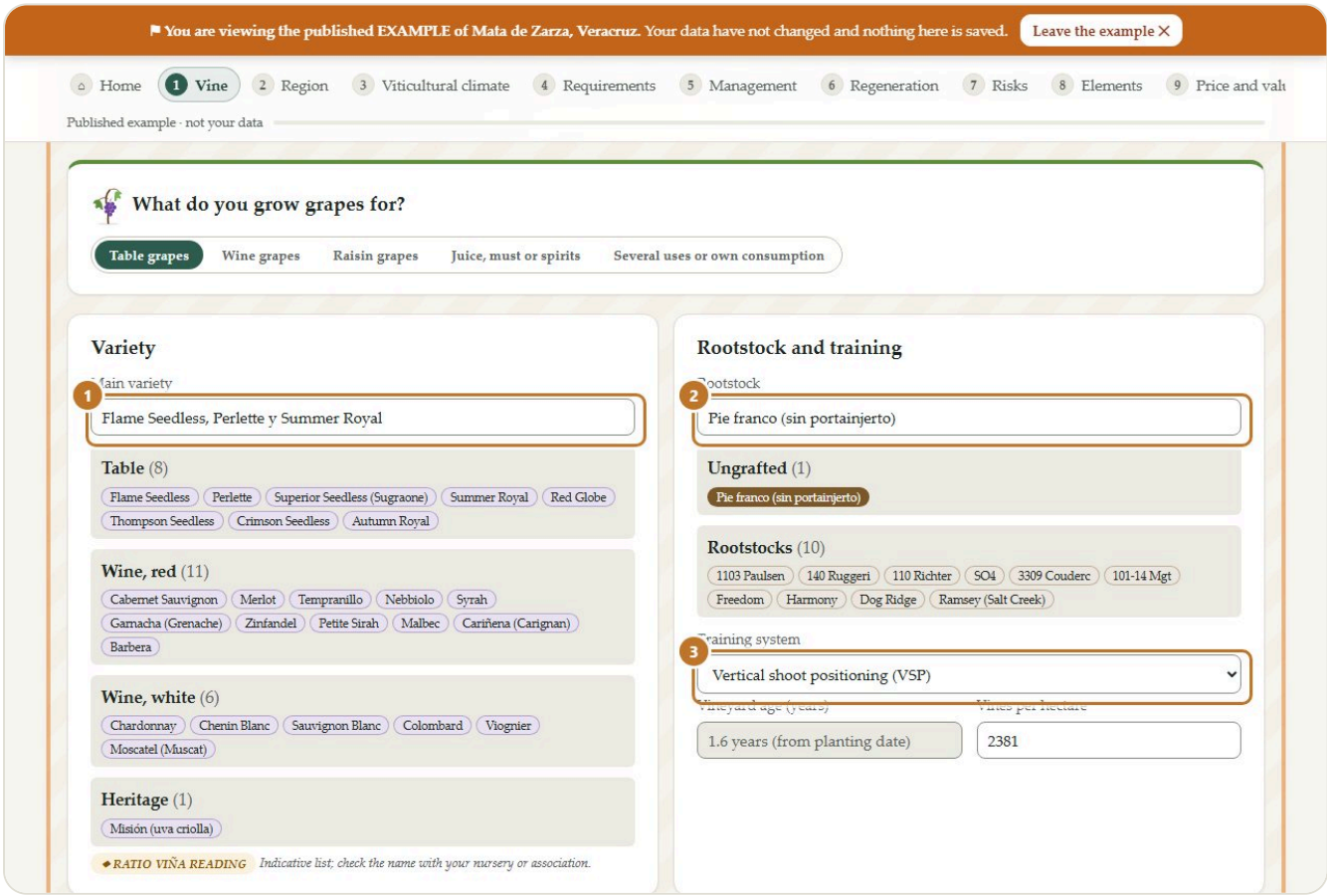


FIGURE 1.1 **Block 1.** The Mata de Zarza case.

1. Main variety.
2. Rootstock.
3. Training system.

✓ **HOW TO WORK OUT VINES PER HECTARE**

Divide 10,000 m² by the in-row spacing times the between-row spacing. With the Mata de Zarza layout, 1.40 m between vines and 3.0 m between rows: 10,000 ÷ (1.40 × 3.0) = **2381** vines per hectare.

1.2 Vineyard history

A vineyard's history explains its present. The **planting date** is set with a knob that runs through the 365 days of the year, plus the year in the box beside it; until you move it, it reads "not entered". From it the app works out the vineyard's **age**. Also choose the **planting material**: unrooted cutting, rooted cutting, nursery-grafted vine or rootstock grafted in the field. Below, in the **pruning record**, add each pruning with its date, type (training, production, summer, second pruning of the year or renewal) and a note. In climates without a marked winter there may be two prunings a year: record them all.

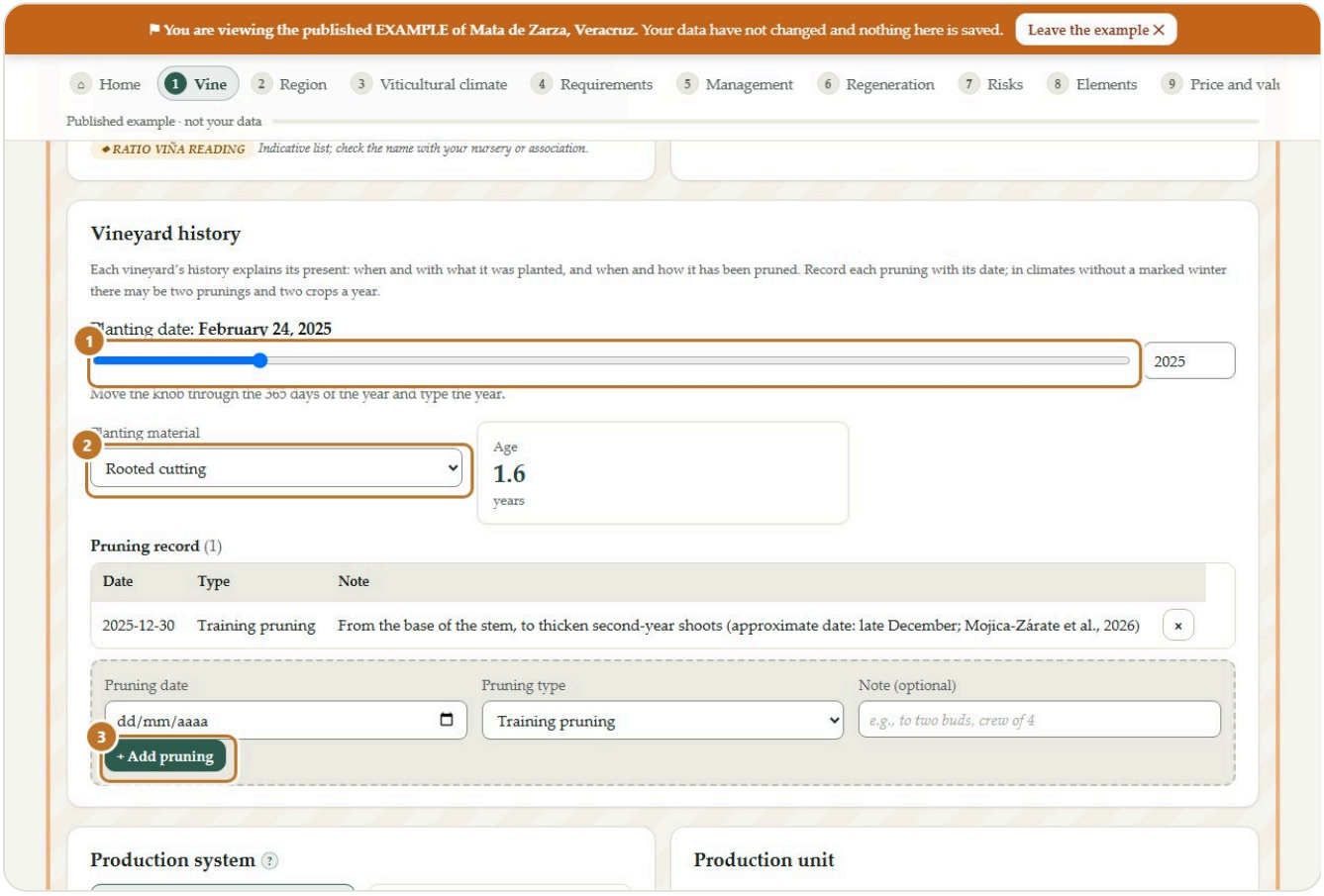


FIGURE 1.2 **Mata de Zarza vineyard history.**

1. Planting-date knob (day of the year) and the year.
2. Planting material.
3. **+ Add pruning**.

◆ IN MATA DE ZARZA

Table grapes of the **Perlette, Flame Seedless and Summer Royal** varieties, introduced as cuttings from vineyards in Hermosillo, Sonora. The vines grow **own-rooted**: the rooted cutting is the whole plant. The planting material is the **rooted cutting**, because cuttings are first rooted in a chamber. Planting took place in **February 2025**; since the guide gives no day, the case uses 24 February, which results from starting rooting in the first ten days of January and adding the 45 days in the chamber. The first pruning, in late December 2025, was a **training pruning from the base of the stem**, to get thicker shoots in the second year (Mojica-Zárate et al., 2026). The production system is "pressurised irrigation" and the training "vertical trellis", because support wires are put up in the first year; both are readings of the guide.

1.3 Production system and unit

The system says how the vineyard is managed: with pressurised irrigation (drip or micro-sprinkler), gravity irrigation, rainfed, organic or transitioning, diversified (with covers, fruit trees or other crops) or a backyard vine; if yours is missing, add it. Then choose the unit type (family, commercial, winery-owned, table-grape ranch, ejido or communal plot, demonstration plot or **Other**), the area and the project name.

◆ IN MATA DE ZARZA

A **1 ha demonstration plot**, one of the 19 cooperating farmers' plots of the first phase (2025).

BLOCK 2 · REGION

2 Region, municipality and locality

Every locality farms its own way. Locate your vineyard and the app brings INEGI's official data.

2.1 From country to locality

Tap your state on the map or pick it from the list; then the **municipality** and the **locality**. The app knows 2,478 municipalities and 54,357 INEGI localities, with their 2020 population, altitude and map point. The region (North, North-central, Central or South) is set automatically. At the start the three fields read "— Choose… —" in grey.



FIGURE 2.1 **Block 2.** Veracruz highlighted; the dot is the locality.

1 Map of Mexico by region.

2.2 Your locality's data



FIGURE 2.2 **INEGI data for the locality.**

◆ **IN MATA DE ZARZA**

Veracruz (South region) › Puente Nacional › **Mata de Zarza**: 372 inhabitants, at 467 m, at 19.230° N and 96.675° W (INEGI, 2020). The guide describes its clay soils and scarce water as the crop's biggest challenges.

✓ **IF YOUR LOCALITY IS MISSING**

Choose the nearest one and note it in the project notes. The catalogue includes localities of 100 inhabitants or more and all municipal seats.

BLOCK 3 · VITICULTURAL CLIMATE

Winkler, Huglin and the nights

3

From each month's temperatures, the app computes three published indices that say how much heat the vine receives and how cool its nights are.

3.1 Monthly temperatures

For each month enter the **maximum**, **minimum** and **mean** temperature in °C. Without the mean, the app takes the average of maximum and minimum. The best data are the **climate normals** of the nearest National Weather Service station with a similar climate; note the source in its field. If there is no nearby station, a reanalysis such as ERA5 gives monthly values for any point in the country, with the caveat that it is not an on-site measurement.

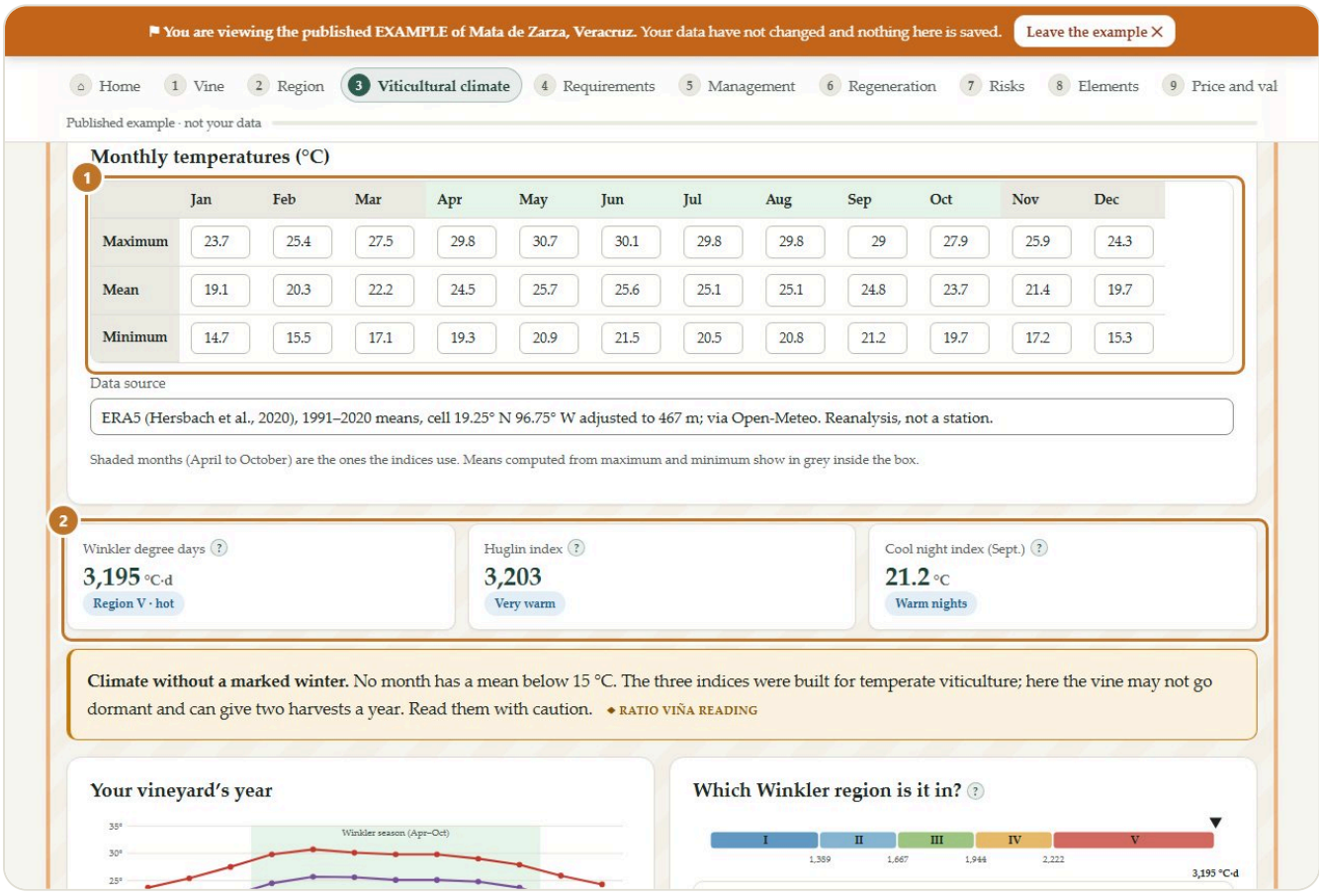


FIGURE 3.1 **Block 3.** Mata de Zarza temperatures and the three indices.

1 Monthly temperature table.

2 The three indices, each with its "?".

3.2 The three indices

INDEX	HOW IT IS OBTAINED	HOW TO READ IT
Winkler degree-days (Amerine and Winkler, 1944)	From April to October, add up how far the mean temperature exceeds 10 °C, times the days of each month.	Five regions: I (under 1,389), II, III, IV and V (over 2,222). More degree-days, riper and less acidic grapes.
Huglin index (Huglin, 1978)	From April to September, add up the average of how far mean and maximum exceed 10 °C, times the days; it weighs sunny hours more.	From "very cool" (1,500 or less) to "very warm" (over 3,000) (Tonietto and Carbonneau, 2004).
Cool-night index (Tonietto and Carbonneau, 2004)	The mean minimum temperature of the ripening month (September).	Very cool (12 °C or less), cool, temperate or warm nights (over 18 °C). Cool nights favour colour and aroma.

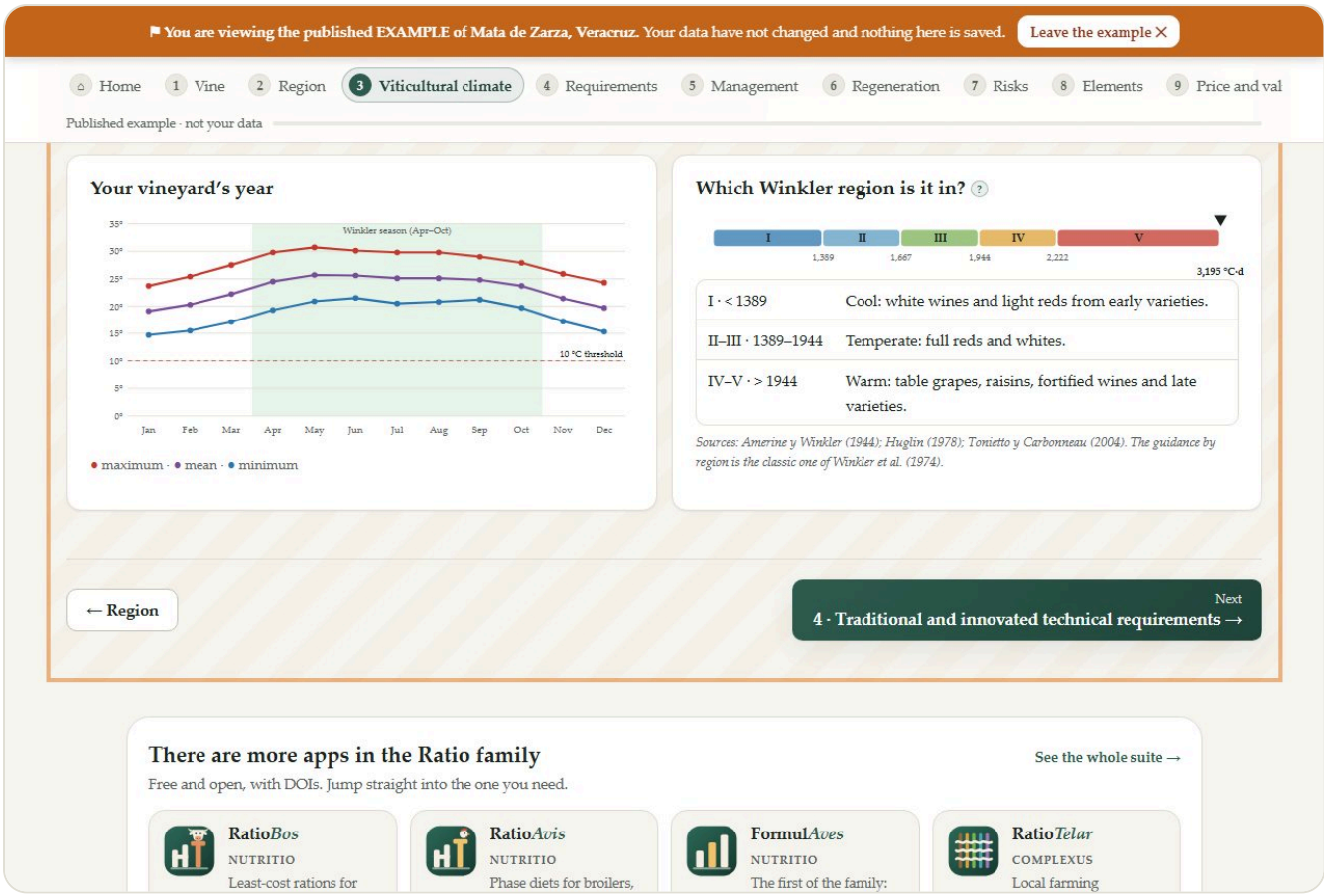


FIGURE 3.2 **Your vineyard's year.** Each month's maximum, mean and minimum, with the 10 °C threshold and the Winkler season.

> **!** **IN CLIMATES WITHOUT WINTER, READ THEM WITH CAUTION**
>
> The three indices were built for temperate viticulture, with an April–October growing season. If no month has a mean below 15 °C, the app shows a warning: the vine may lack a marked rest and its cycle may not match those months. In the tropics table grapes call for short-cycle varieties and techniques different from those of temperate zones (Kok, 2014).

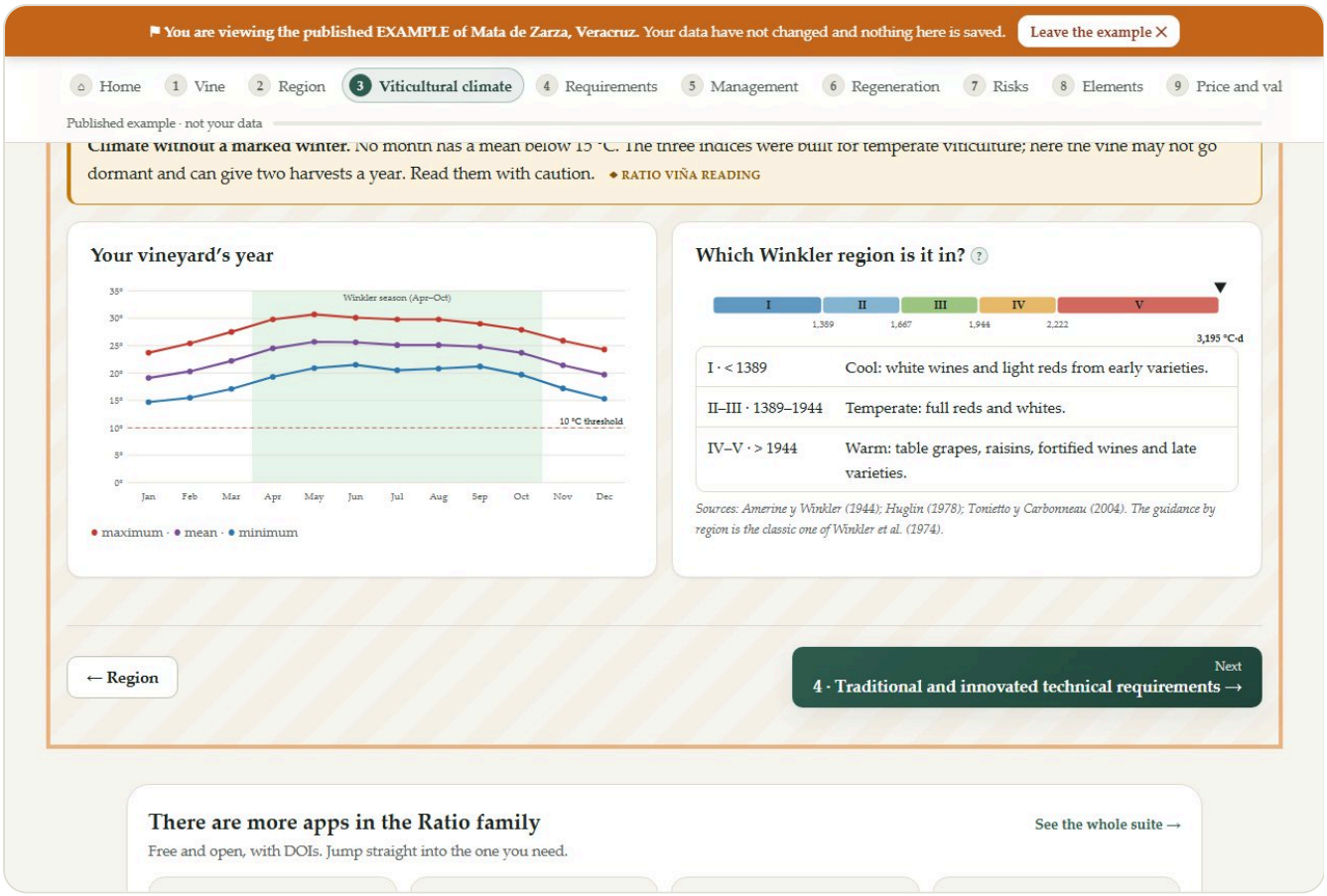


FIGURE 3.3 **Which Winkler region is it in?** The full scale and where the vineyard falls.

◆ **IN MATA DE ZARZA**

The guide publishes no temperatures and there is no weather station in the locality; nor is the Huatusco Regional Academic Centre station suitable, as it has another climate type. So the case uses the 1991–2020 monthly means of the **ERA5** reanalysis (Hersbach et al., 2020) in the cell at 19.25° N and 96.75° W, adjusted to 467 m. The coolest month is January, averaging 19.1 °C, and the warmest May, at 25.7 °C. With those data: **3,195 Winkler degree-days (region V)**, **Huglin index of 3,203 (very warm)** and **21.2 °C cool-night index (warm nights)**. As no month drops below 15 °C, the no-marked-winter warning appears: in Mata de Zarza the indices serve for comparison, not for choosing varieties as in Sonora or Baja California.

BLOCK 4 · REQUIREMENTS

4 What the vine needs to do well

Seven requirements, one per stage of the cycle, and those your vineyard has added over the years.

4.1 The baseline questionnaire's requirements

Each requirement sums up what a stage of the cycle needs: healthy, well-rooted material, planted in moist soil and protected in year one; suitable pruning and a sound trellis; enough rest, or management to break dormancy where chilling is lacking; well-drained soil and timely irrigation; plant health with scouting; balance between foliage and clusters; and harvest at the ripeness the use requires. For each, mark whether it is met: **Yes**, **Partly** or **No**. As there is no published grapevine Stack for a producing vineyard yet, the last six are a ◆ Ratio Viña reading of viticulture texts (Winkler et al., 1974; Keller, 2020); the establishment one comes from the Mata de Zarza guide.

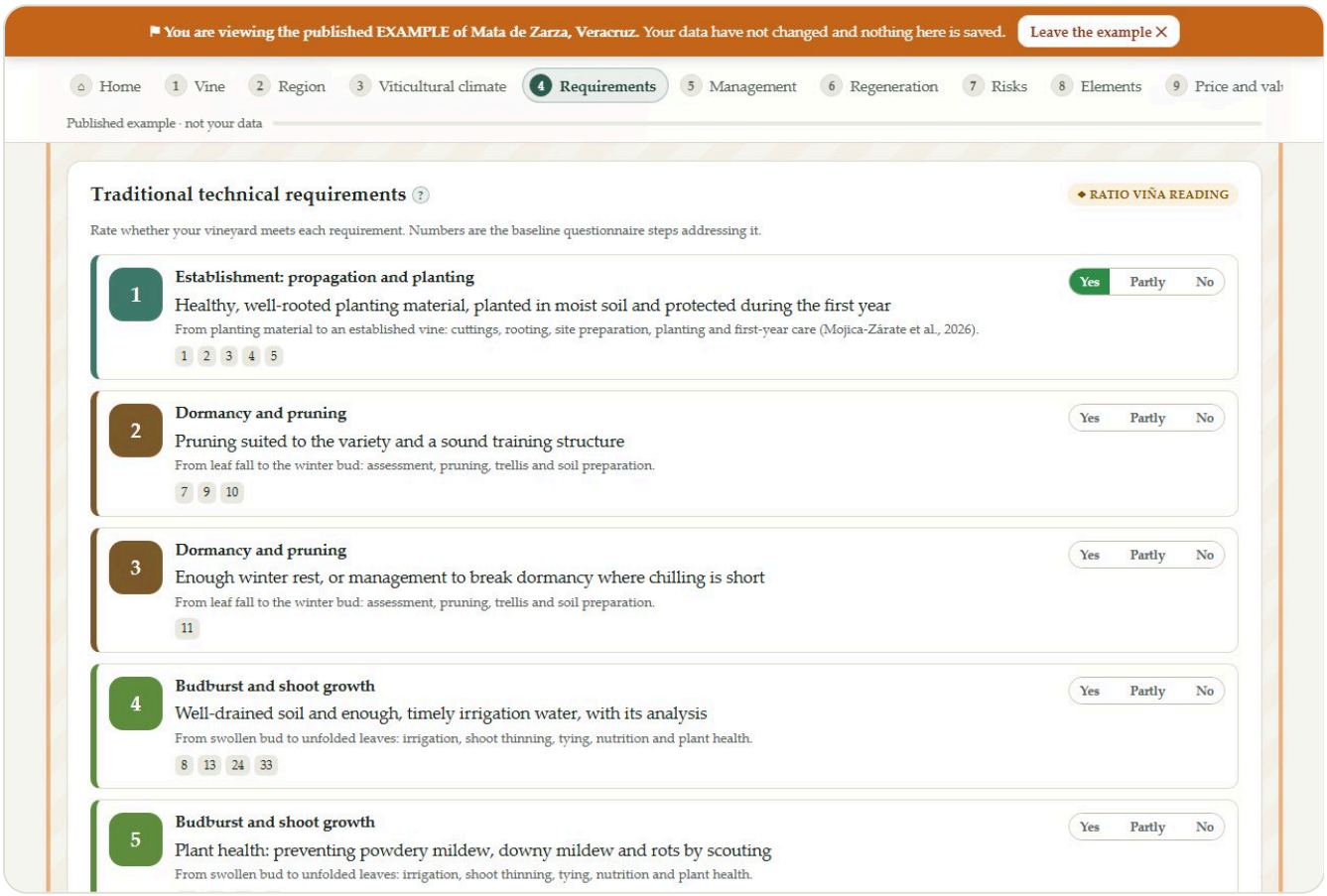


FIGURE 4.1 **Requirements.** With their stage and the Yes, Partly or No traffic light.

4.2 Innovated requirements

Below you add what your vineyard has learnt: write the requirement, choose its stage, a description and the source (farmer, adviser, research, programme), and tap **Add requirement**.

◆ **IN MATA DE ZARZA**

The establishment requirement is marked **Yes**: the guide describes selecting healthy cuttings, planting in freshly watered holes and first-year protection. The other six stay unmarked, because the guide covers up to preparation for pruning and the plot has not yet flowered. Two innovated requirements from the guide are added: **clean, sieved river sand** for the rooting chamber, "not industrial or building sand", and **water for constant irrigation in the first three months** after planting.

✓ **ALWAYS RECORD THE SOURCE**

"Said by the cooperating farmer, 2025" or "UACh practical guide, 2026" is enough. The source is printed in the report and gives your record its worth.

BLOCK 5 · MANAGEMENT

5 From cutting to harvest

The heart of the system: what is done, in what order and when. Everything that follows is computed from these activities.

5.1 Seven stages, 37 steps and 67 activities

The baseline questionnaire orders vineyard management into seven stages. The first, **Establishment: propagation and planting**, has 6 steps and 15 activities, from choosing the site and planting material to training the young vine, taken from the Mata de Zarza guide (Mojica-Zárate et al., 2026). The other six follow the annual cycle by the grapevine's BBCH phenological stages (Lorenz et al., 1995): dormancy and pruning; budburst and shoot growth; flowering and fruit set; berry development and veraison; ripening and harvest; and post-harvest, destination and balance. No activity comes answered. At the top you see how many you answered, how many you practise (they enter the network) and how many not; the coloured buttons filter steps by stage.



FIGURE 5.1 **Block 5.**

1 Filter by stage.

5.2 Inside a step

Opening a step shows its activities. For each one answer **Yes, I do** or **No**; those left "not answered", with a striped background, do not enter the network. **All yes** or **All no** answers the whole step. Check the **coloured squares**, which show whether the activity adds (green) or subtracts (red) for soil, water, atmosphere, vegetation and fauna, and correct them if your case differs; set the **dates** of the main activity; and with **+ Innovative activity** add what your vineyard does differently, choosing the elements involved.

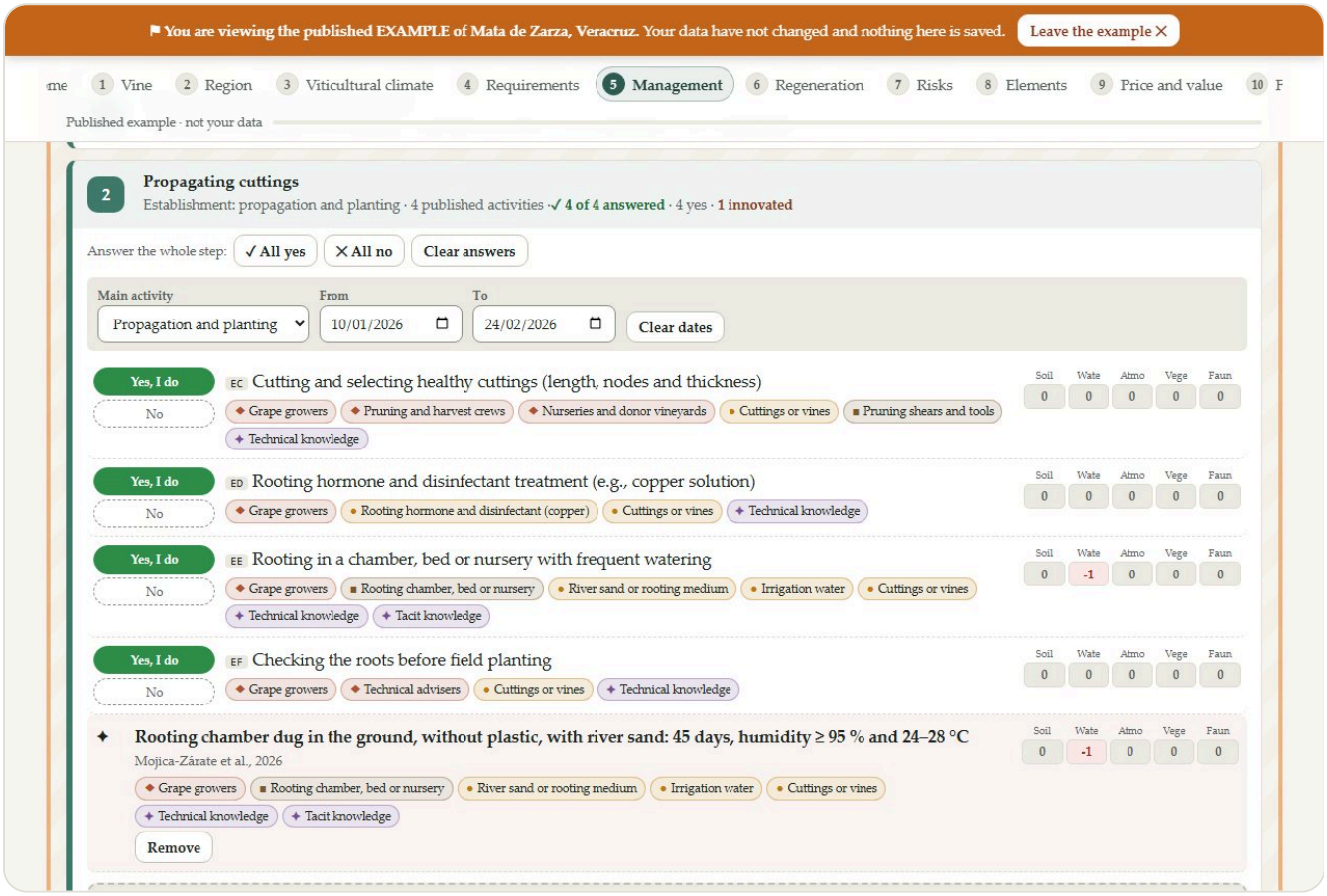


FIGURE 5.2 **Step 2, propagating cuttings.** The four activities answered "yes", their dates and the Mata de Zarza rooting chamber as an innovative activity.

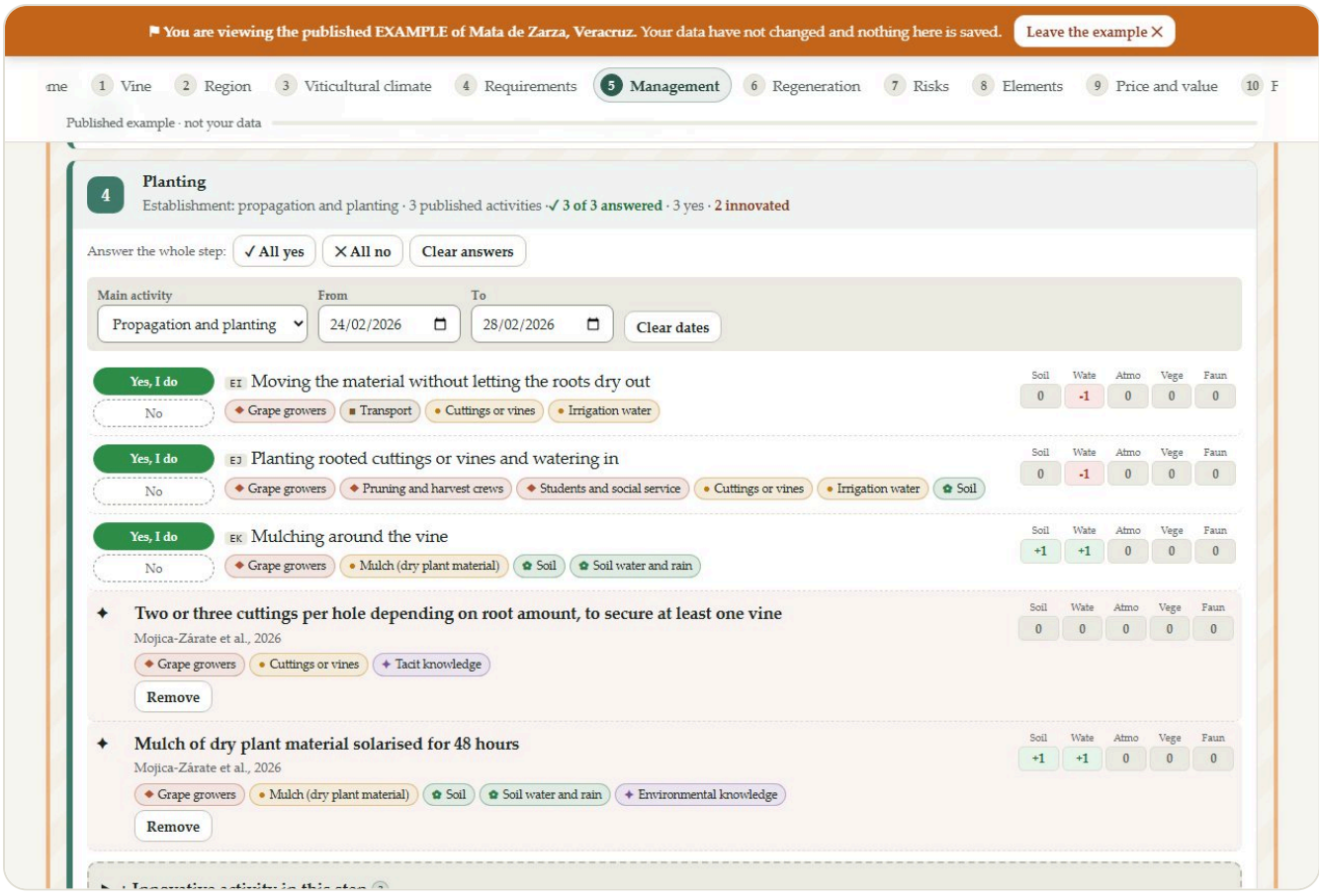


FIGURE 5.3 **Step 4, planting.** With two innovations from the guide: several cuttings per hole and solarised mulch.

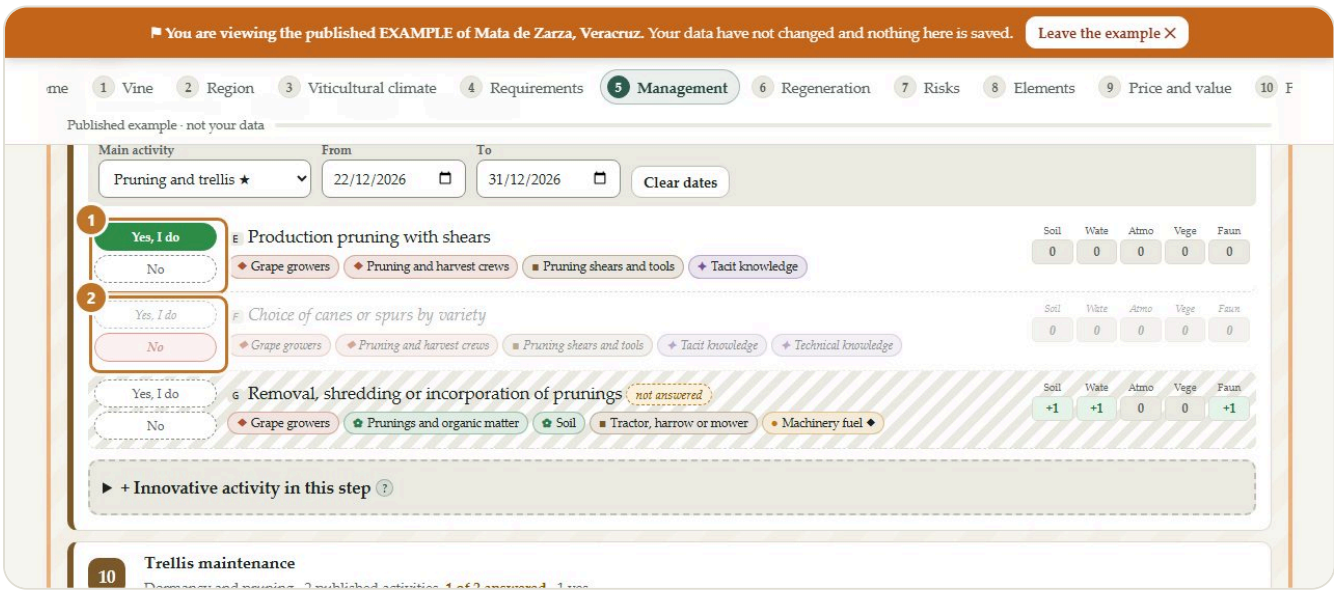


FIGURE 5.4 **Yes / No answers.** In Mata de Zarza's winter pruning, pruning with shears is "yes" and choosing canes or spurs is "no", because the first-year vine is pruned from the base.

> **?** **WHY DO EACH ACTIVITY'S ELEMENTS MATTER?**
>
> Because that is how the network is woven: two elements taking part in the same activity become linked. If the rooting chamber uses "river sand", "irrigation water" and "tacit knowledge", that practice strengthens the link between them.

5.3 The filling guide and the published case

Every block shows the **"Not sure how to fill it in?"** box. Tapping **See the filling guide** opens a window, with the "GUIDE · REFERENCE ONLY" watermark, explaining what to enter, where to find the data and how to fill it in. From there you can enter the Mata de Zarza case with **See the published Mata de Zarza case**. When you close the window your fields stay the same.

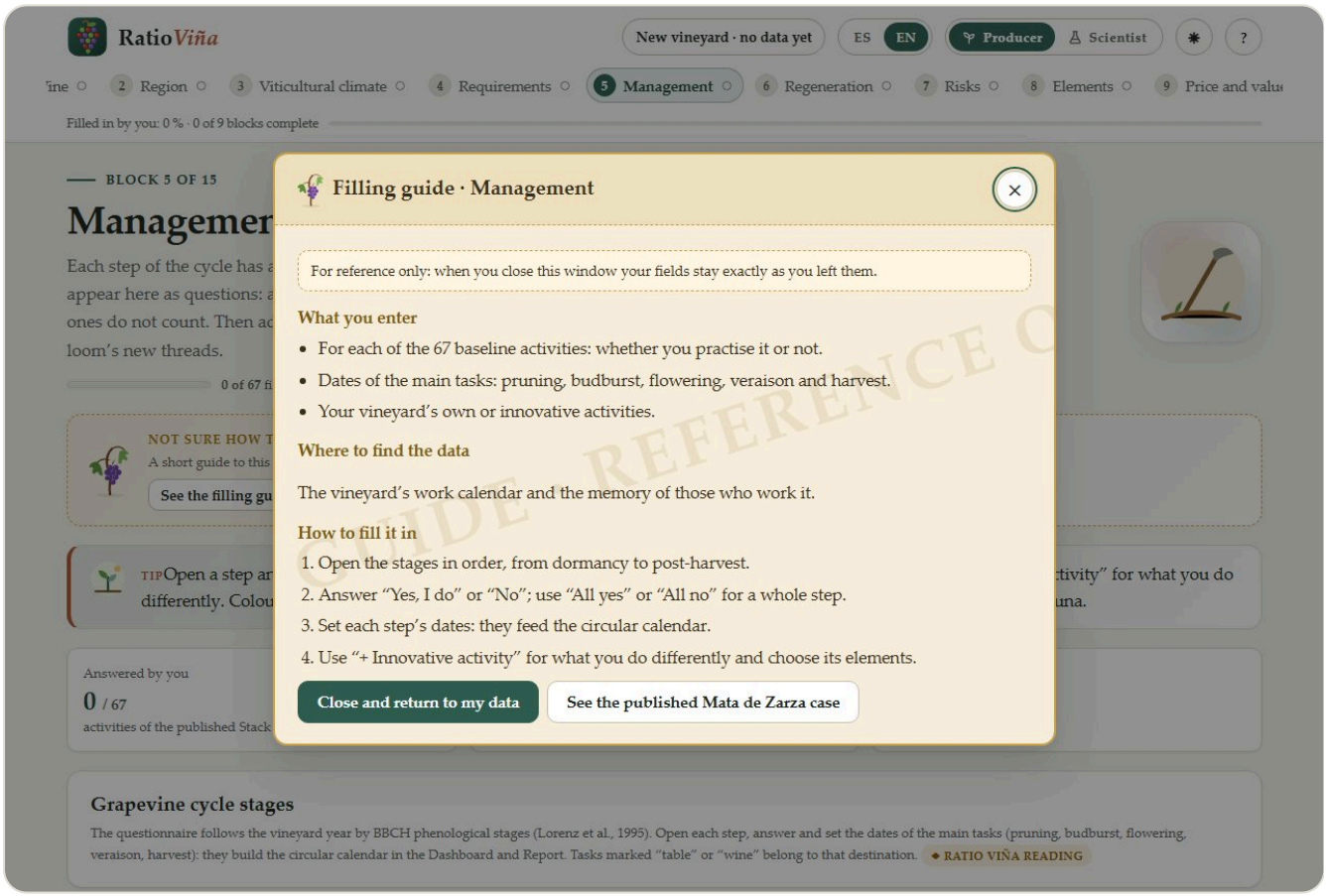


FIGURE 5.5 **Filling guide** for Block 5.

IN MATA DE ZARZA

Of the 67 activities, **48 are answered**: 27 “yes” and 21 “no”; the other 19 stay unanswered because the guide does not mention them. The **15 establishment activities** are “yes”, with their dates: cutting cuttings in the first days of January, 45 days in the rooting chamber at 24–28 °C and 95 % humidity or more, holes 20 cm wide and 70 cm deep at 1.40 × 3.0 m, planting with 1.2 L per hour for 40 hours and then 3 L per vine a day, mulching, protection against leaf-cutter ants and animals, lifting onto supports and support wires. From the annual cycle, “yes” goes to winter pruning, wire checks, compost or manure (chicken manure), irrigation, tying, fertilising with diammonium phosphate (40 g per vine), weeding with a brush cutter, irrigation by demand, bio-inputs (mountain microorganisms), recording costs in logbooks, teaching students and farmers and exchange with Sonora growers. Flowering, thinning, girdling, veraison, harvest, packing and sales are “no”: the plot has not reached those stages. There are also **8 innovative activities** from the guide, such as the rooting chamber dug in the ground, without plastic, with river sand, and protecting each vine with net sacks held by stakes.

BLOCK 6 · REGENERATION

6 Is your vineyard regenerating or wearing out?

Eight 0–4 knobs measure whether practices give back to soil, water and biodiversity more than they take.

6.1 The regeneration index

The gauge sums up the eight knobs in a 0–100 number and a class: **degrading** (under 25), **conventional** (25–49), **regenerative transition** (50–74) and **regenerative** (75 or more). Next to it is the **regeneration profile**, a radar that shows at a glance which principle is strong and which weak.

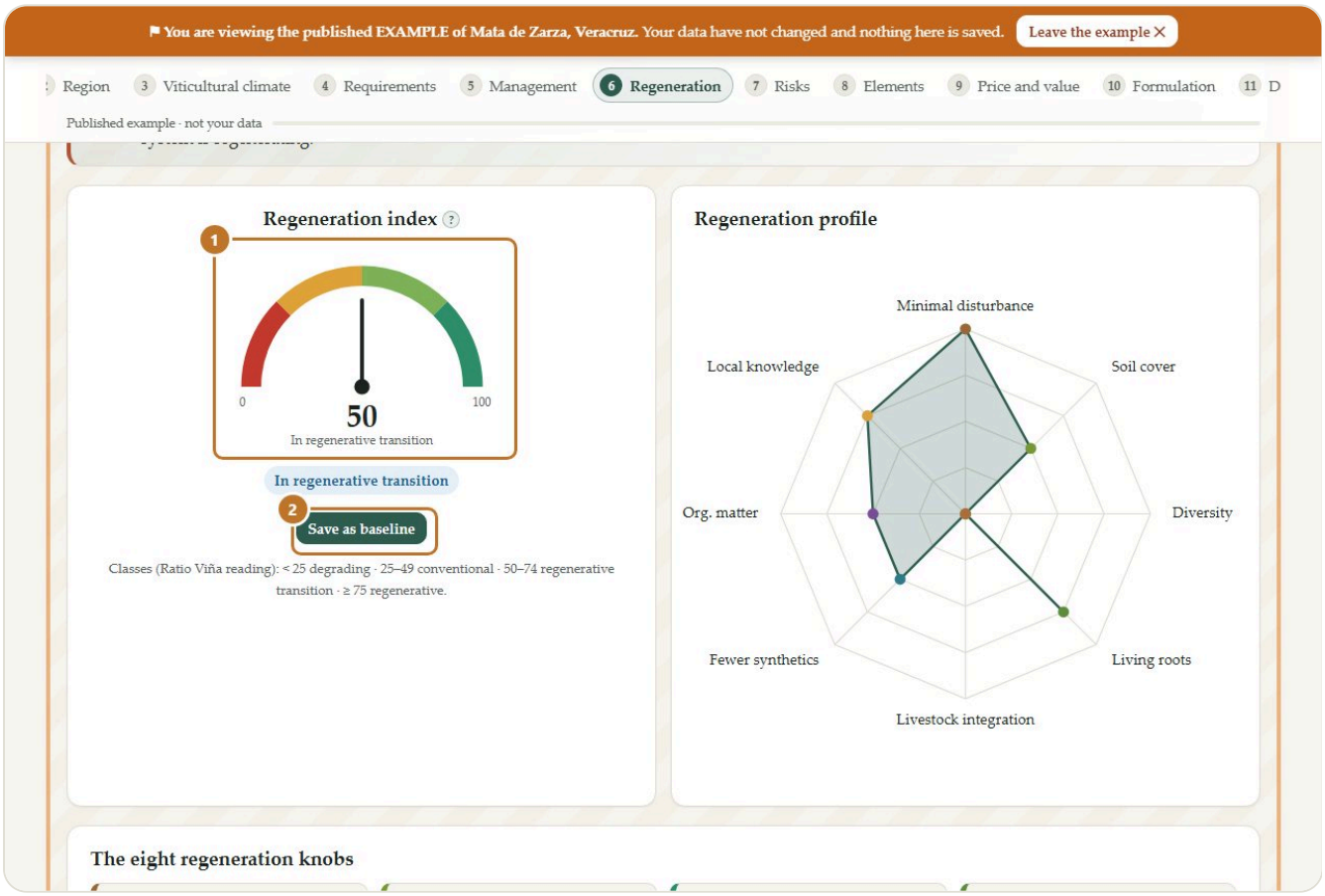


FIGURE 6.1 **Regeneration index and profile.**

1 0–100 gauge, with its class.

2 **Save as baseline**.

6.2 The eight knobs

Each knob answers one question and brings a **suggested value** computed from your Block 5 activities, with its reason. Until you confirm it, it looks **dashed and grey**, labelled “Suggested, not confirmed”. If the value fits your vineyard, tap Accept suggestion; if not, move it. It then reads “✓ Confirmed by you”. While knobs are missing, the index is flagged **provisional**.

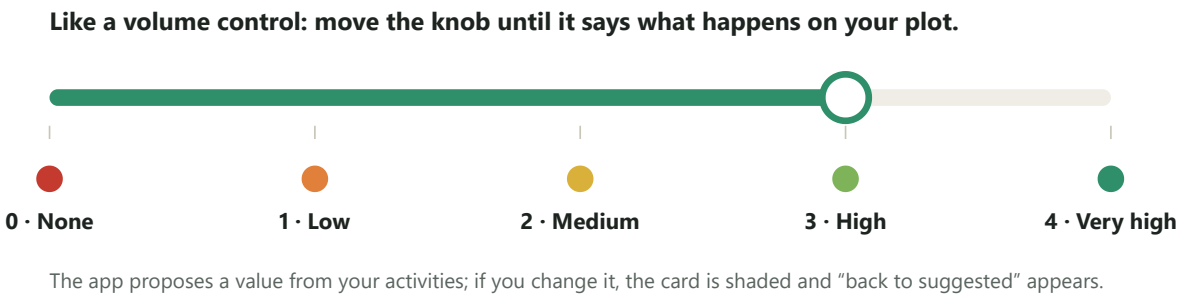


FIGURE 6.2 **The knob scale.** Regeneration and value use ordinal scales: each point has a clear meaning, not a made-up number.

KNOB	QUESTION	MATA DE ZARZA
Minimal tillage	How many machinery or plough passes move the soil each cycle?	4
Soil cover	How many months is it covered by crop, prunings, mulch or green manure?	2
Species diversity	How many species are grown or raised together?	0
Living roots	How many months are there living plants on the plot?	3
Animals integrated	Are there animals grazing, providing manure or living in the unit?	0
Fewer agrochemicals	What share of inputs is synthetic versus bio-inputs?	2
Organic matter	Are residues returned to the soil and organic matter measured?	2
Local knowledge	How much does management rely on tacit and cultural knowledge and the family?	3

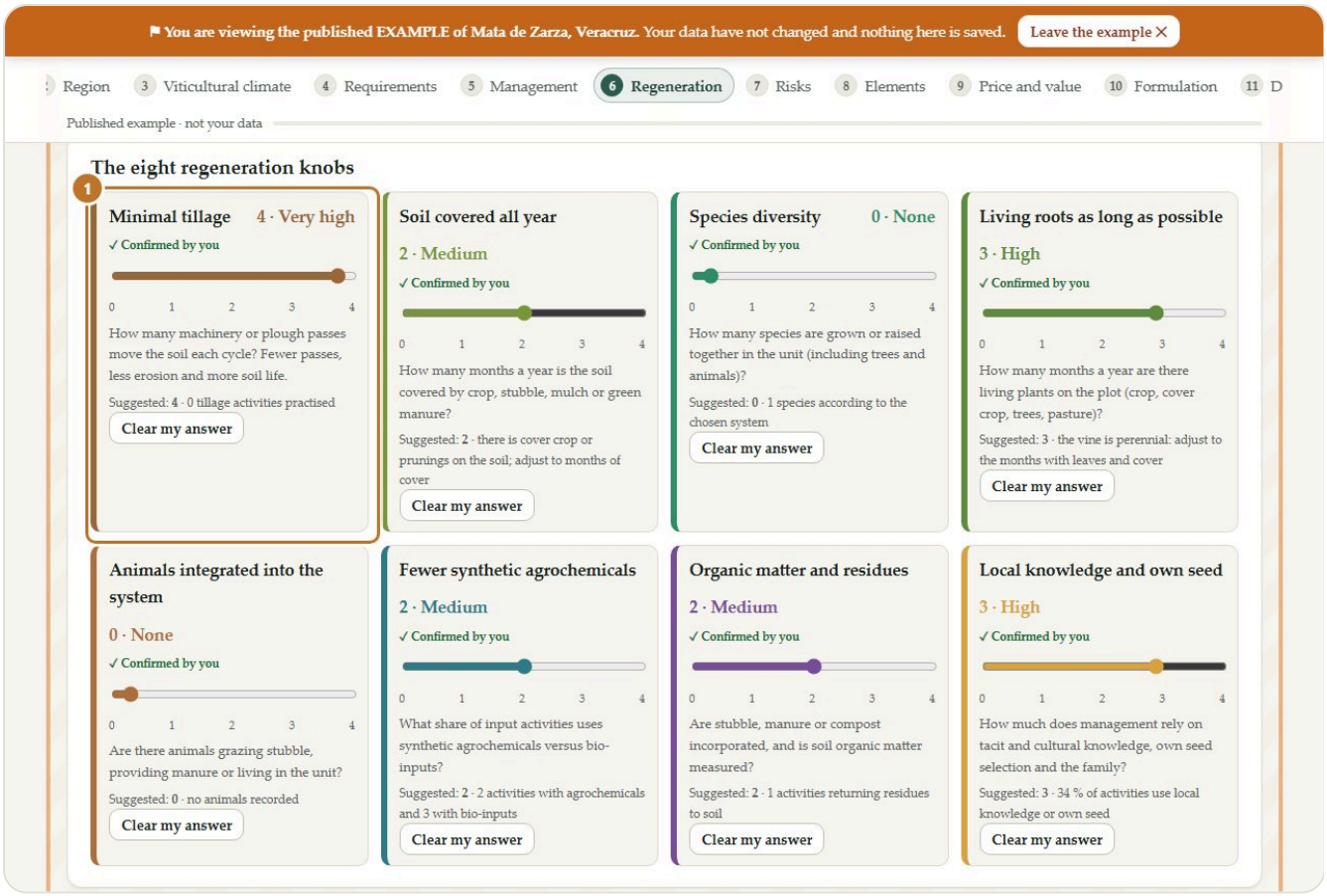


FIGURE 6.3 **The knobs.** Each with its question and suggested value.

1 Minimal tillage: 4 · Very high, because no activity in the case moves the soil with machinery.

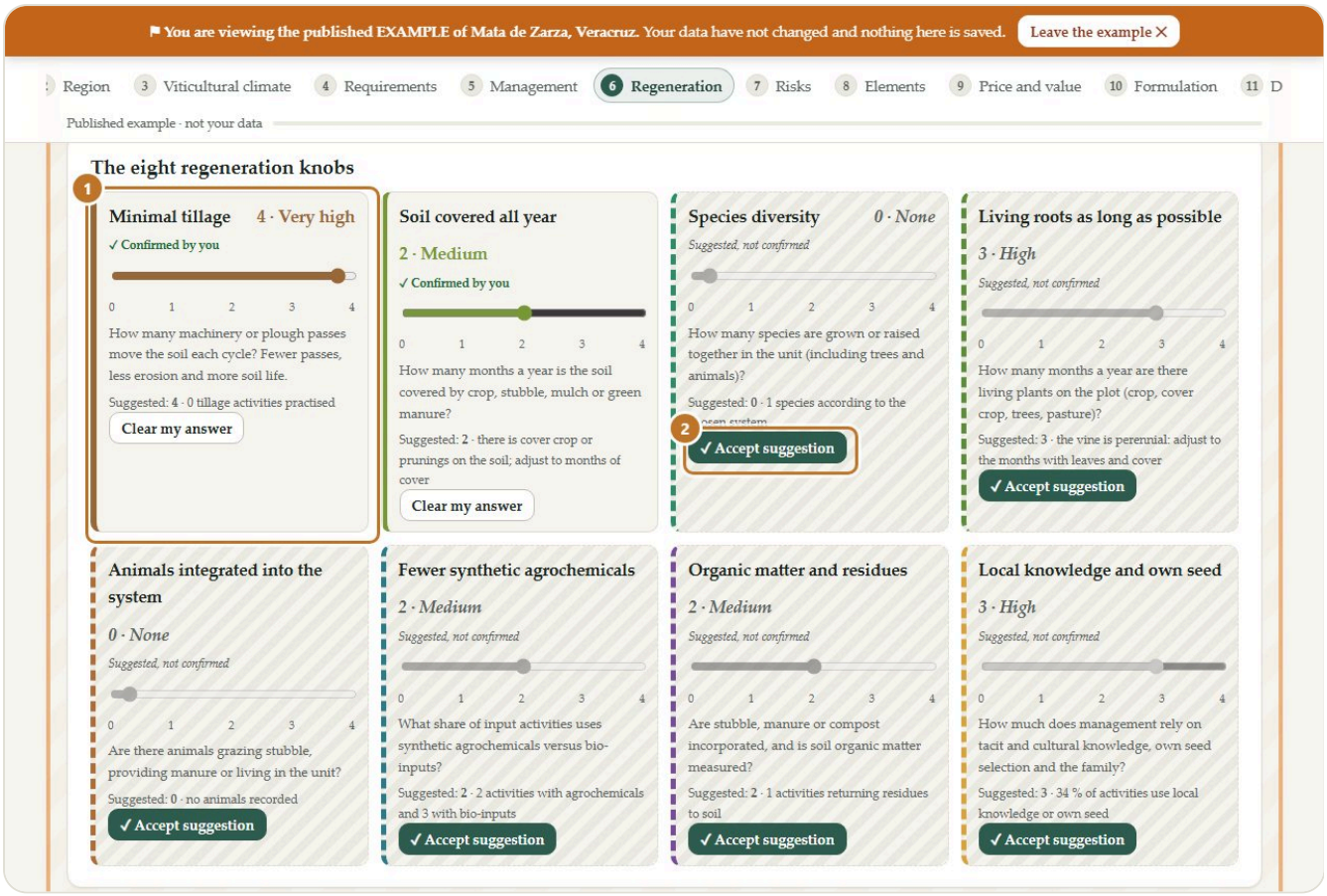


FIGURE 6.4 **Confirmed and suggested.** The first two knobs are already confirmed; the others are still dashed, with their **Accept suggestion** button.

◆ **IN MATA DE ZARZA**

Accepting the suggested values, the index is **50 · Regenerative transition**. It is raised by no activity moving the soil with machinery, the vine being perennial, mulch and chicken manure, and a third of the activities (34 %) relying on tacit and cultural knowledge. It is lowered by the plot having a single species and no animals. On inputs there are 2 activities with synthetic agrochemicals (diammonium phosphate and fertigation) and 3 with bio-inputs or organic manures, including mountain microorganisms. The most regenerating activities are compost or manure and mulching; those subtracting most are the irrigation ones, because of the water drawn.

6.3 Baseline and organic matter

Regeneration shows over time. Tap **Save as baseline** today and come back next cycle: the app says whether the index rose ("the system is regenerating") or fell ("the system is wearing out") and on which knobs. If you have two soil tests, enter the % organic matter before and now: it is the most direct proof.

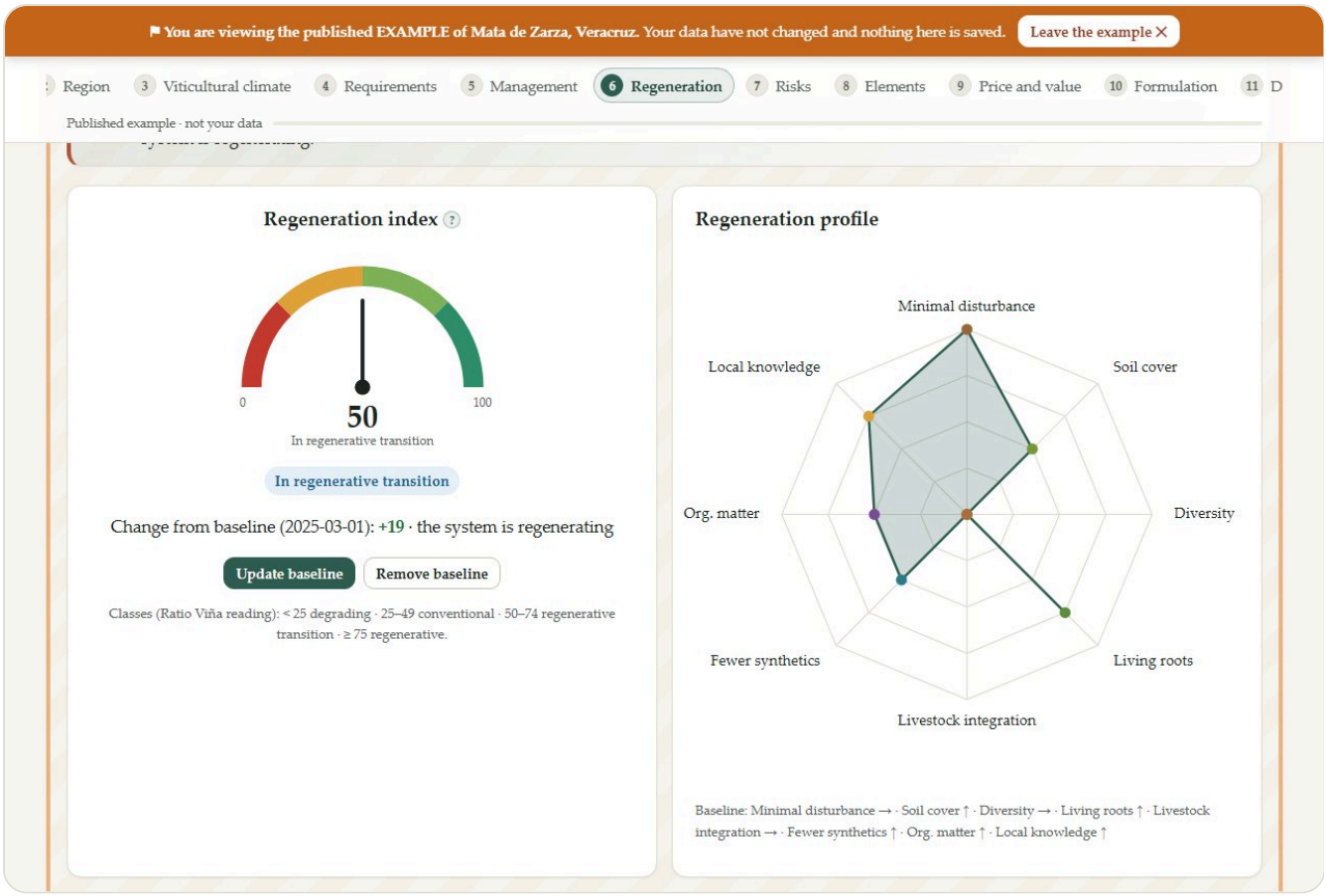


FIGURE 6.5 **Comparison with a baseline** (illustrative, unpublished example: baseline of 31 in March 2025 and organic matter from 1.6 to 1.8 %).

→ **DECISION RULE · WHERE DO I START?**

Soil cover at 0 or 1	Leave chopped prunings between rows or mulch the row; record it as an activity.
Fewer agrochemicals at 0 or 1	Replace one application with a bio-input and record it as an innovative activity.
Diversity or animals at 0	If you sow a cover between rows or bring in backyard poultry, record it as an activity and adjust those knobs; protect young vines from the hens.

BLOCK 7 · RISKS

7 What could go wrong and how serious it would be

Twenty-eight hazards in a likelihood-by-impact matrix, reduced by your vineyard's adaptive capacity.

7.1 The index and the matrix

At the top you see the **risk index** (0 to 100), the **adaptive capacity** and what it is made of, how many risks are high or very high, and which is the highest. Below, the **risk matrix**: each circle shows how many risks fall in that cell.

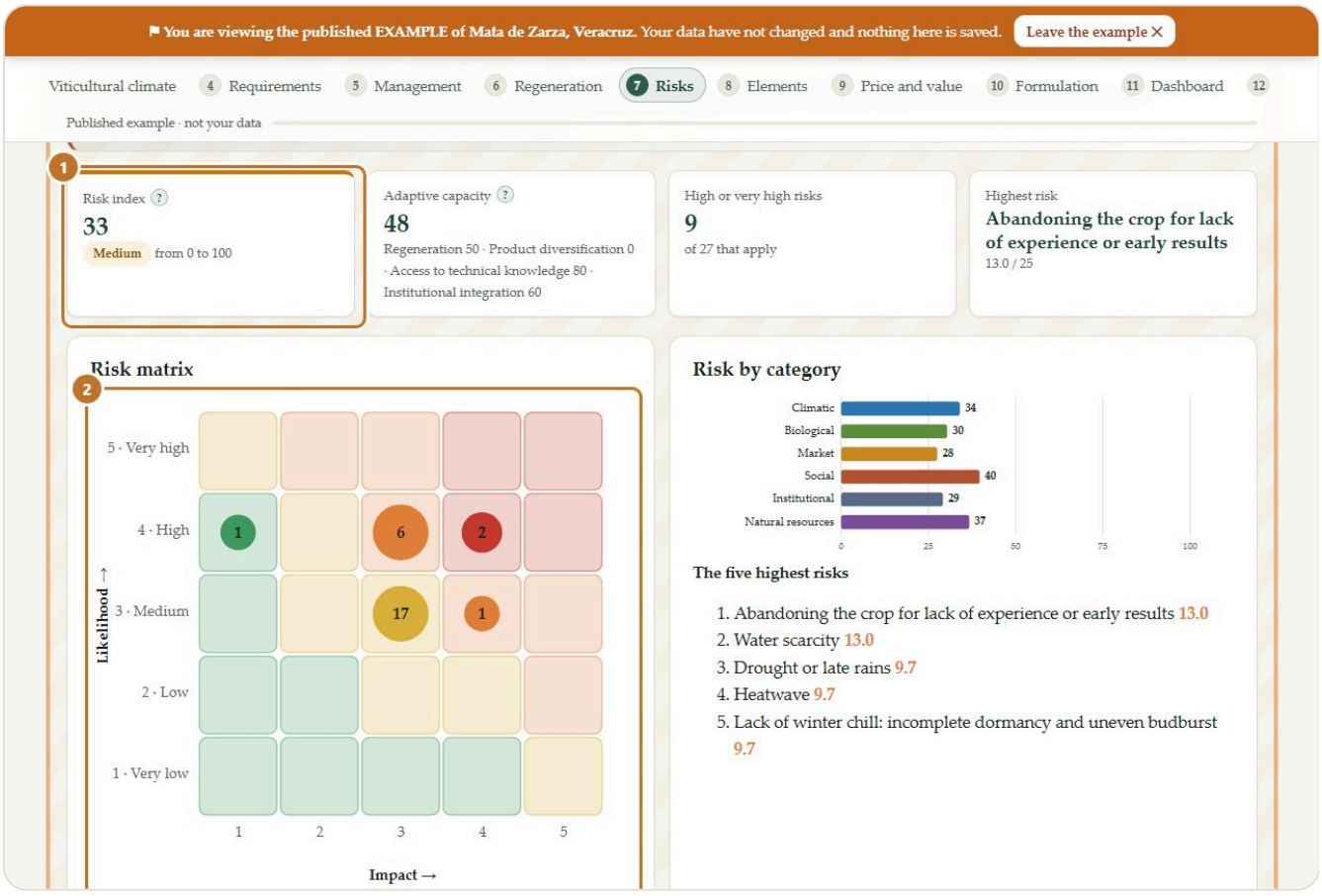


FIGURE 7.1 **Block 7.**

1. Risk index and its class.
2. Likelihood-by-impact matrix.

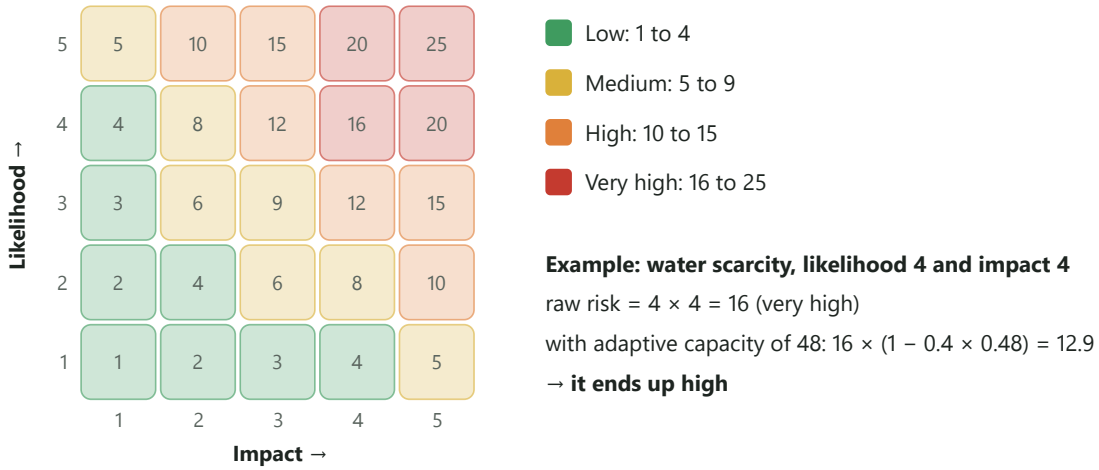


FIGURE 7.2 **How to read the matrix.** Each cell multiplies likelihood by impact. Your system's adaptive capacity and the controls you already have lower the risk. The example is water scarcity in the Mata de Zarza case.

7.2 The 28 hazards

They are grouped into **climatic** (drought or late rains, frost, hail, heavy rain, heatwave, cyclones, lack of winter chill), **biological** (pests, fungal diseases, Pierce's disease, leaf-cutter ants, animal damage, diseased or poorly rooted material, herbicide drift, weeds), **market** (price fall, input cost rise, new crop without a market, middlemen), **social** (labour shortage and succession, abandoning the crop in the first years, loss of local knowledge, insecurity), **institutional** (support withdrawal, land tenure) and **natural resources** (erosion, water scarcity, clay or poorly drained soils). Each new hazard shows its source. For each: untick it if it does not apply, and move **likelihood**, **impact** and the **control** you already have.

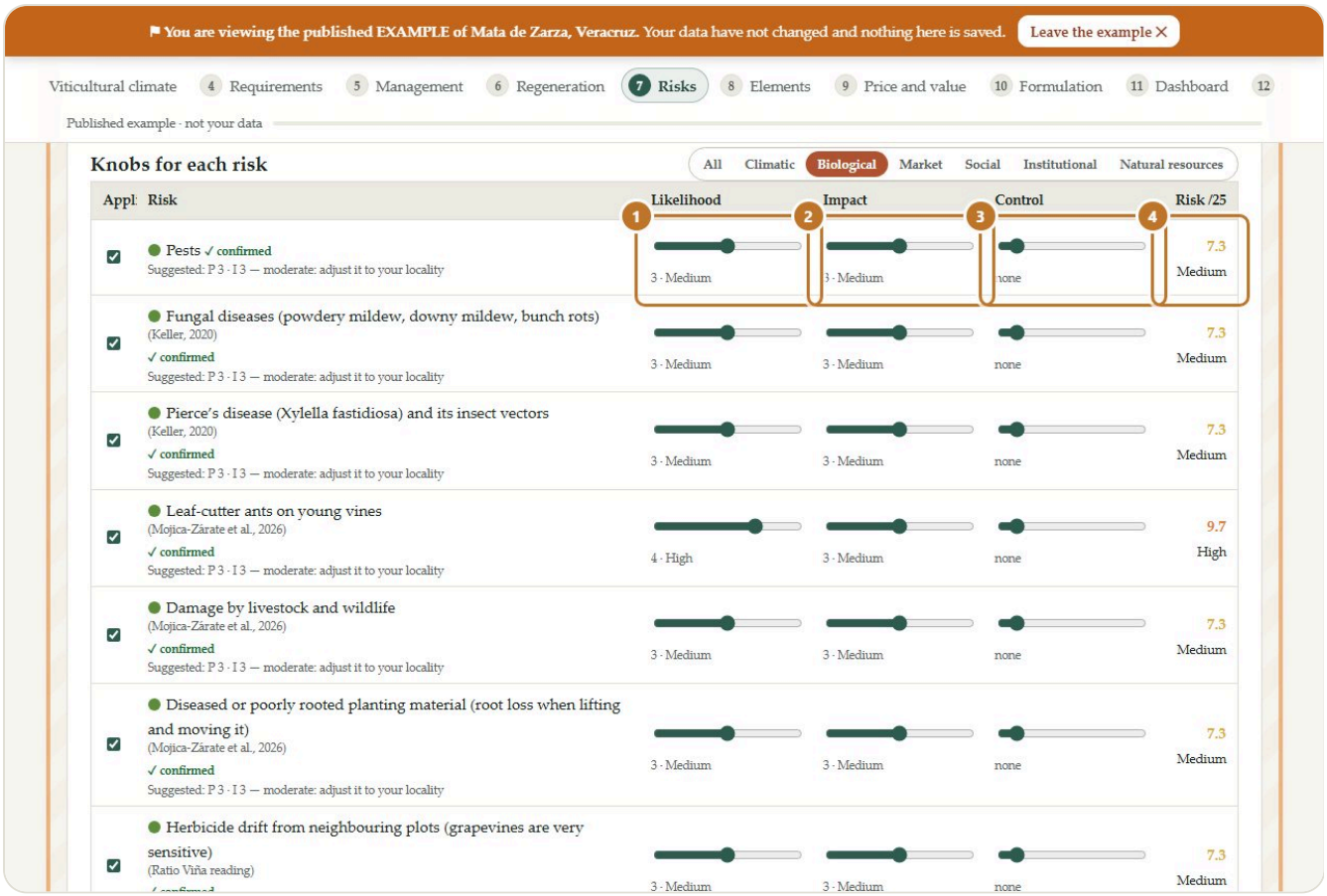


FIGURE 7.3 **Knobs of the biological hazards.**

1. Likelihood: how often it happens (1 very low to 5 very high).
2. Impact: how much damage it does.
3. Control you already have: none, low, medium or high.
4. Resulting risk, 0 to 25, with its level.

> **?** **WHERE DO THE SUGGESTED VALUES COME FROM?**
>
> From your data and the outlook. For example, drought and heatwave likelihood rises to 4 because the trend scenario assumes 0.2 °C of warming per decade (IPCC, 2018); the impact of input cost rises depends on what share of your activities uses purchased inputs. What cannot be deduced stays at 3, "moderate: adjust it to your locality".

◆ **IN MATA DE ZARZA**

Risk index **33 · Medium**, with adaptive capacity of **48** (regeneration 50, product diversification 0, access to technical knowledge 80, institutional integration 60). There are 9 high risks among the 27 that apply; frost was unticked because, with the ERA5 data, the lowest monthly mean minimum is 14.7 °C. The highest, at 13.0 out of 25, are **water scarcity** and **abandoning the crop in the first years**: the guide singles out water as the biggest difficulty, and its Table 1 records 19 cooperating farmers in 2025 and 4 in 2026. Next, at 9.7, come drought, heatwave, lack of winter chill and **leaf-cutter ants**, which damaged leaves in the first three weeks. The values of the hazards the guide describes are a ◆ Ratio Viña reading of its text: each plot's farmers adjust them.

BLOCK 8 · ELEMENTS

Who and what takes part

8

The loom's threads: actors, institutions, inputs, equipment, natural and financial resources, knowledge, policies, disturbances and products.

8.1 Ten categories and 75 elements

Each element belongs to one of ten categories and takes part in one or more activities. The base catalogue has 75 elements of a Mexican vineyard; the establishment stage added, among others, nurseries and donor vineyards, students, cuttings, rooting hormone, river sand, mulch, the rooting chamber, guards and stakes, and leaf-cutter ants. The bar next to each element shows how many activities it appears in. If one from your vineyard is missing, add it with its name and category and choose it in the activities where it takes part.

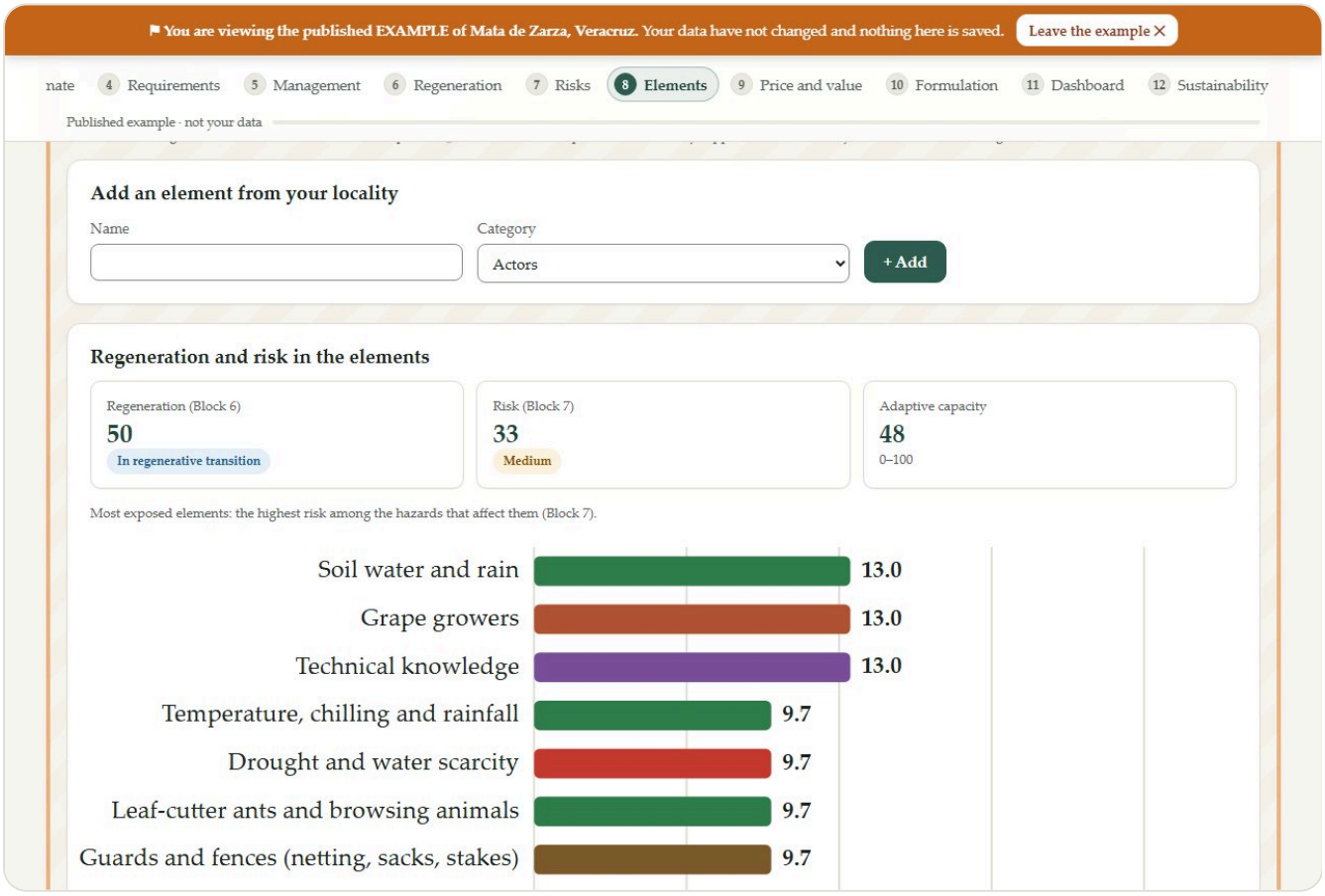


FIGURE 8.1 **Vineyard elements** by category.

> **!** **AN ELEMENT WITH NO ACTIVITY STAYS OUT OF THE NETWORK**
>
> If you add “Farmers’ savings fund”, then choose it in the activity where it takes part (Block 5, “+ Innovative activity”). Otherwise it stays in the catalogue but is not woven in.

8.2 Elements exposed to risk

At the end of the block, the **Regeneration and risk in the elements** card shows which elements are most exposed: the highest risk among the hazards affecting them (Block 7).

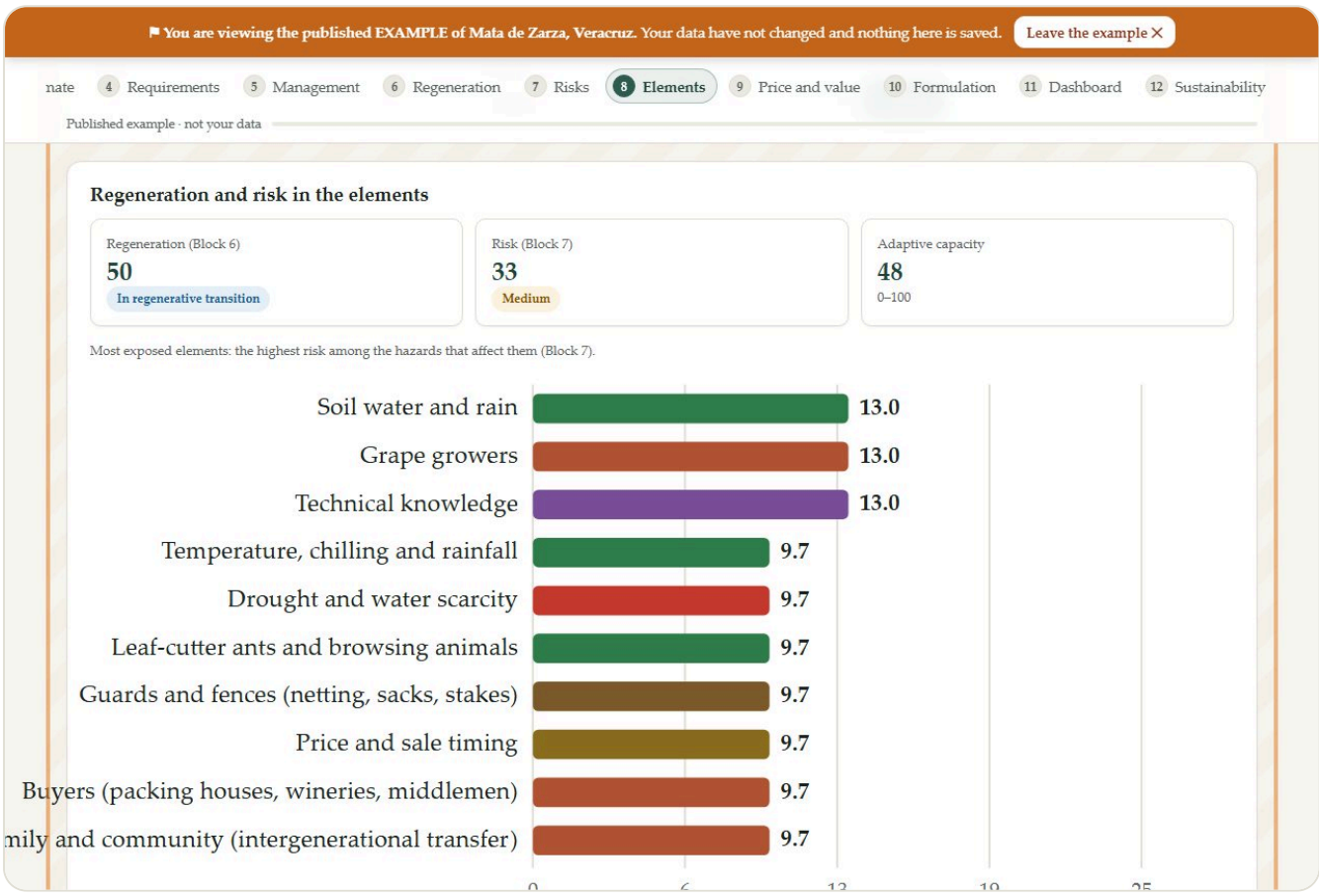


FIGURE 8.2 **Most exposed elements** in Mata de Zarza.

> **◆** **IN MATA DE ZARZA**
>
> The 35 practised activities use **40 elements**. The guide documents, in its Table 1, the 2025 actors: 26 invited and 19 cooperating farmers from Mata de Zarza, 32 students and 1 PROFONI student from UACh, 4 lecturer-researchers from the Huatusco Regional Academic Centre, 22 cutting harvesters and 2 planting-material suppliers from Hermosillo, Sonora, and 1 input supplier from Atotonilco el Alto, Jalisco: 107 people from four states.

BLOCK 9 · PRICE AND VALUE

9 What it costs and what it is worth

Price is what the market charges; value is the element's importance to the system. They are not the same.

9.1 Price per hectare

Choose a category with the tabs. For each element enter the **unit price**, the **unit** (cutting, kg, L, day's labour) and the **quantity per hectare**; the app computes $/ha. **Financial resources** holds production cost, income and sale price: they feed the margin in Sarandón's index and net income in the outlook.

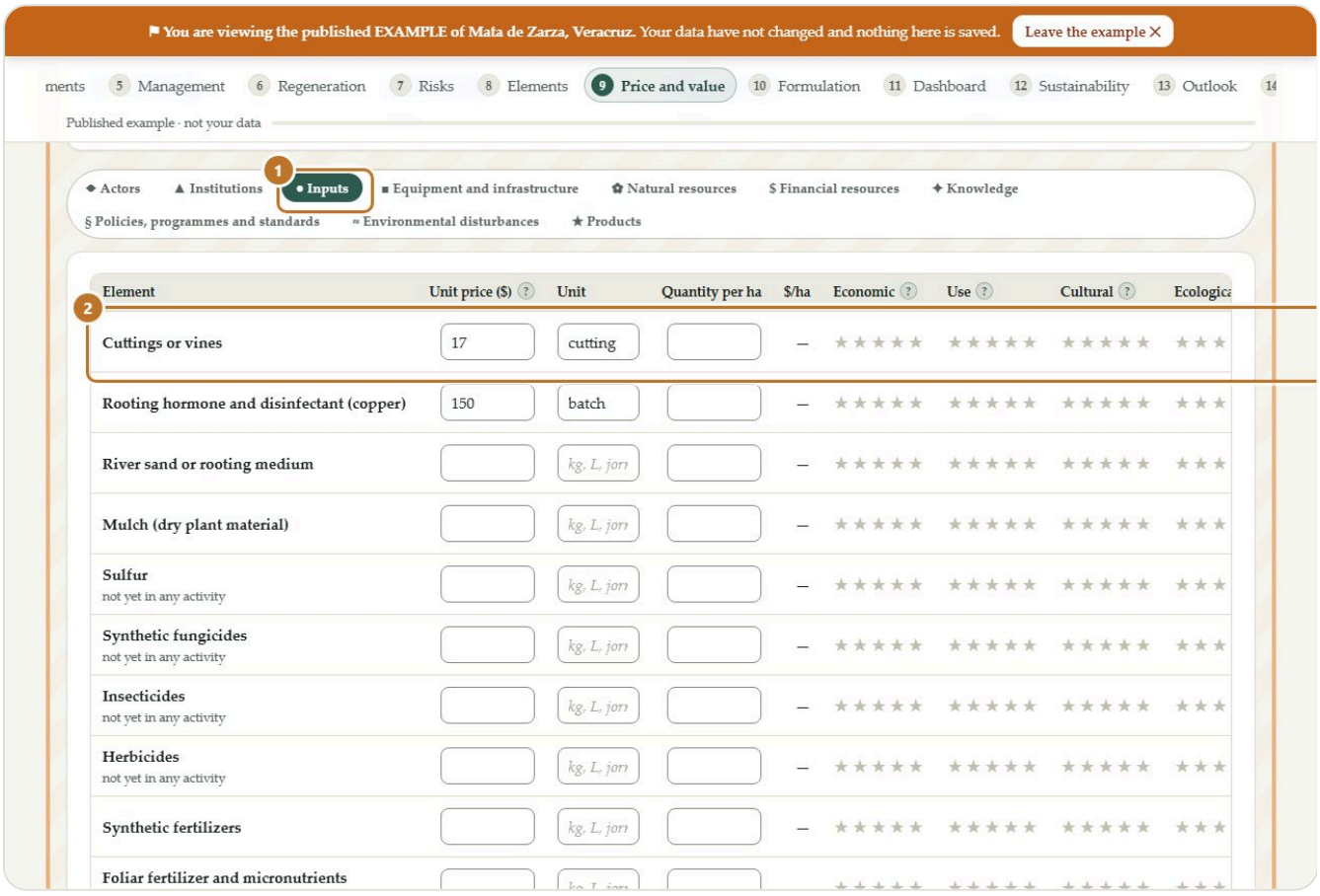


FIGURE 9.1 **Inputs with their unit prices** in the Mata de Zarza case.

1. Category tab.
2. Price, unit, quantity per hectare and the four value dimensions.

9.2 Value in four dimensions

Rate each element from 0 to 5 on **economic**, **use**, **cultural** and **ecological** value. At the top, the **weight** knobs say how much each dimension counts. If you do not rate, the app uses the **structural value**: how

much the element takes part in the activities.

◆ **IN MATA DE ZARZA**

The guide's Table 2 publishes establishment costs from the field logbooks: cuttings at MXN 17 each, transport at MXN 6.10 per cutting, labour days at MXN 400 and rooting hormone at MXN 150; with 3,600 cuttings, establishment totals MXN 119,160. For years 2 and 3 it projects MXN 444,691 of maintenance from FIRA data, an expected yield of 17 t/ha according to Sonora growers, expected income of MXN 867,000 and a benefit/cost ratio of 1.5. The case enters only the **unit prices**; it leaves cost and income per hectare without data because the table combines the cost of all 19 plots with a per-hectare projection from another region. So margin and net income await each plot's figures.

9.3 Your catalogue as CSV

With **Download catalogue (CSV)** you save your elements, prices and values to a spreadsheet; with **Load catalogue (CSV)** you bring them into another project.

BLOCK 10 · FORMULATION

10 Weaving the system

Elements become nodes, shared activities become links and value sets each thread's thickness.

10.1 The "Weave the system" button

Tap **Weave the system**. The app builds the network and takes you to the Dashboard. Next to the button you see how many nodes, links and activities your system has and its density. Below, the **system model**: the environment (climate and disturbances), agents, knowledge Stack, management, inputs, resources, institutions, policies and products, with their arrows.



FIGURE 10.1 **Block 10.** In Mata de Zarza: 42 nodes, 175 links, 35 activities and density 0.203.

1 **Weave the system**; once woven it reads "Re-weave the system".

✓ **RE-WEAVE WHENEVER YOU CHANGE SOMETHING**

If you add activities, elements or values, come back to this block and re-weave. The Dashboard warns with an amber box if you have not woven yet.

10.2 The 3D system model

Below the button is the **three-dimensional system model**. Layers run from bottom to top: the environment (climate and disturbances); inputs (natural resources, inputs, equipment and finance); agents, knowledge and institutions; management, with the 5B, their steps and activities; and outputs (products, market and well-being). Above sits the halo of emergence and adaptation. Each dot is an element of your network and each line a link; dashed lines are the cycle feedbacks.

Drag to rotate it (on a phone, with your finger); zoom with + and −, Ctrl + wheel or pinching, and use Front, Top or Reset. Tap an element to see only its neighbours and tap it again to go back. Visible links shows only the strongest links or all of them. The 2D sketch tab keeps the box diagram.

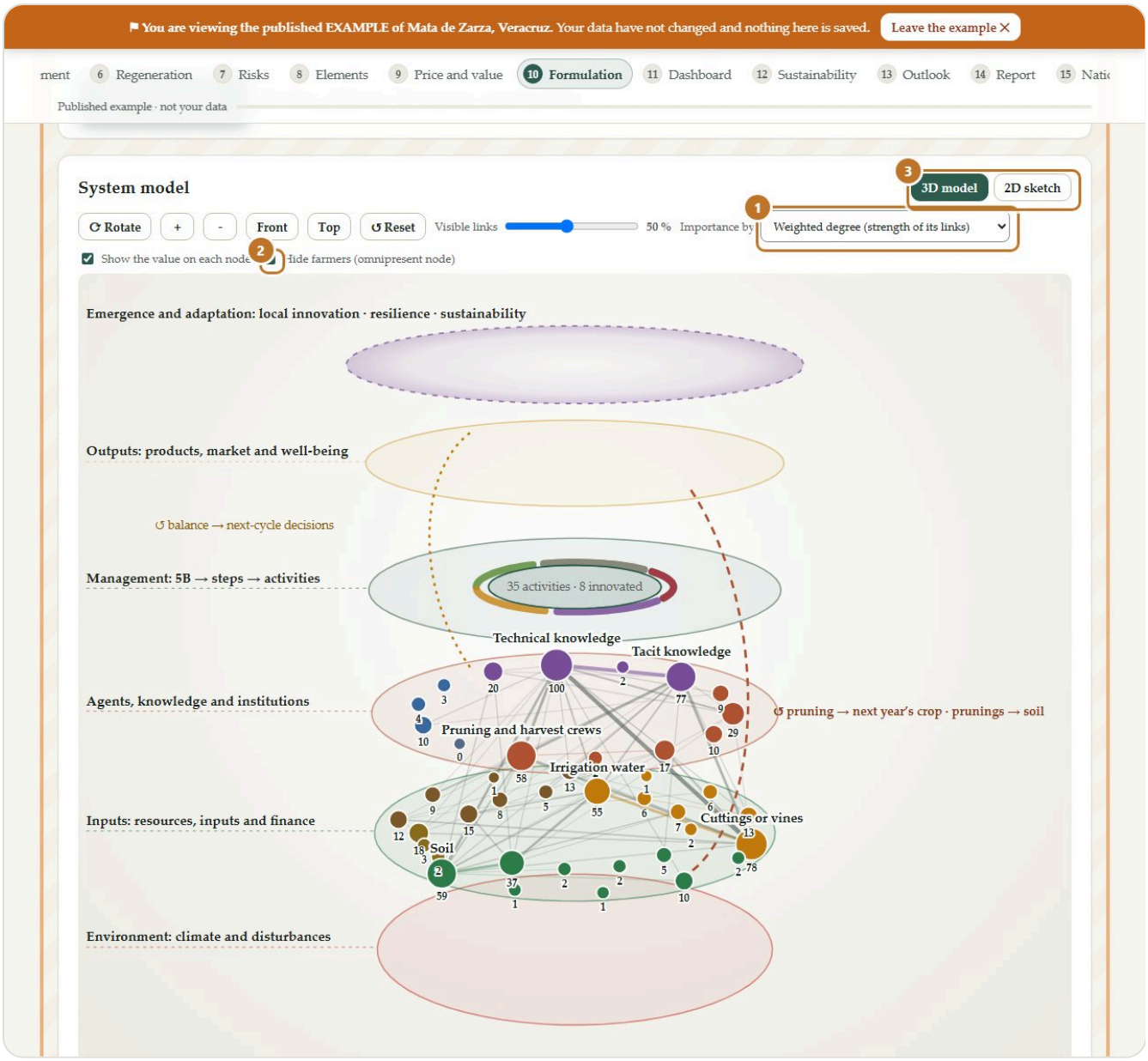


FIGURE 10.2 **3D system model** in Mata de Zarza.

1. **Importance by**: the measure behind each element's size and number.
2. Hide growers: on by default, because they take part in almost every activity.
3. **3D model** and **2D sketch** tabs.

10.3 The importance of each element

The number next to each element is its **importance, from 0 to 100**: the most important element in your network scores 100 and the rest are measured against it. Choose the measure in **Importance by**: weighted degree (how much it interacts with the others; the default), betweenness (how much it bridges parts of the system), eigenvector (whether it is linked to the important elements) or element value. Below the model is the list of the 10 most important elements; tap one to see it in the model.

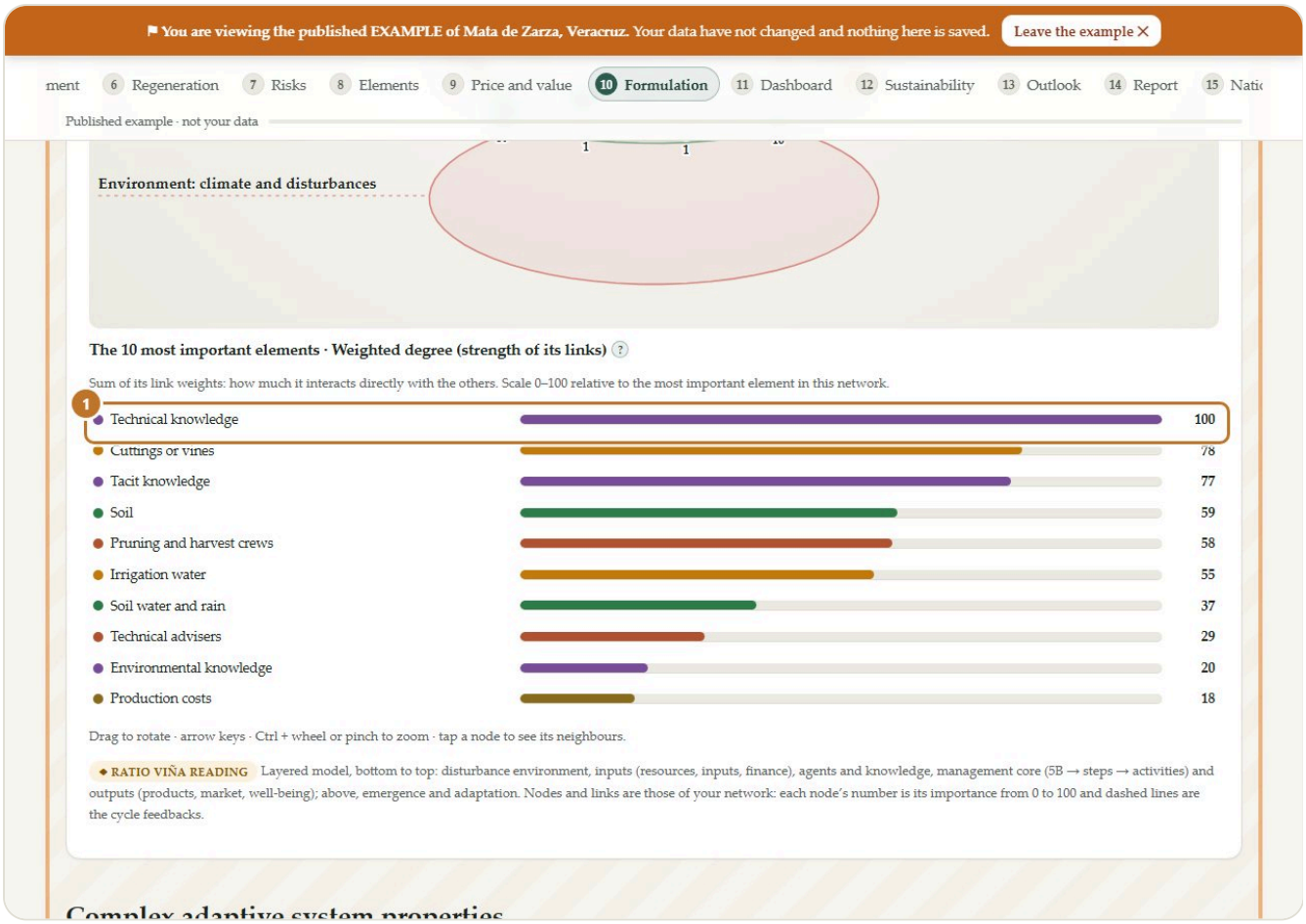


FIGURE 10.3 **The 10 most important elements** in Mata de Zarza, by weighted degree.

1 The most important element scores 100.

◆ **MATA DE ZARZA**

By weighted degree, the most important elements are technical knowledge (100), cuttings or vines (78), tacit knowledge (77), soil (59) and pruning and harvest crews (58).

BLOCK 11 · DASHBOARD

11 Reading the whole system

Who holds the system together, which threads weigh most, which groups form, who is close to whom and when each thing happens.

11.1 The summary at the top

The top indicators give the network size (nodes and links), its density, clustering and mean path, modularity Q and mean sustainability. Below, the **System regeneration and risk** card brings the indices from blocks 6 and 7, the radar and the matrix.

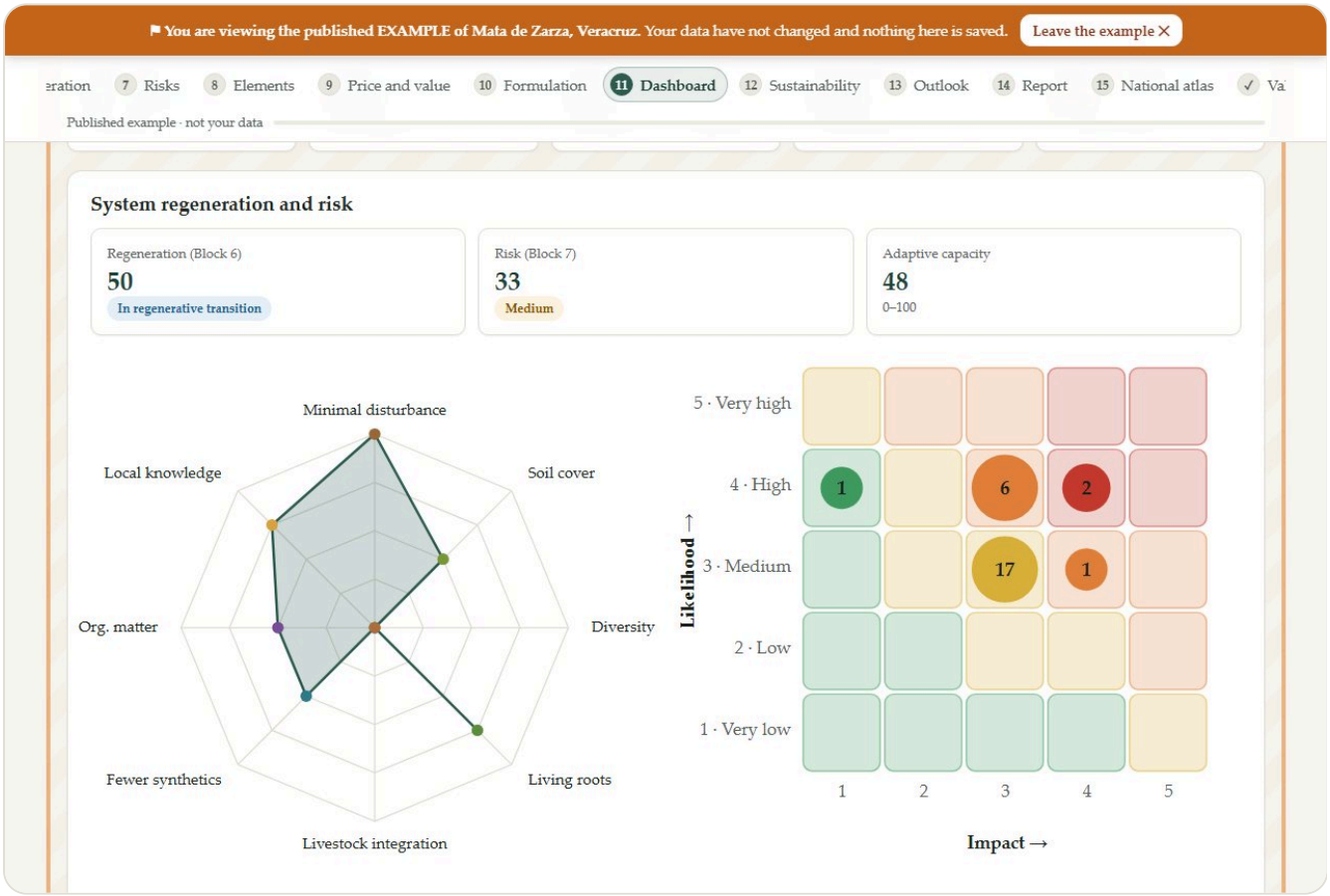


FIGURE 11.1 **The Mata de Zarza Dashboard.**

11.2 The system network

Each dot is an element, coloured by category; its size depends on the measure you choose at the top: **weighted degree** (how related it is), **betweenness** (how much it bridges), **eigenvector** (whether it relates

to the important ones) or **closeness**. Tap a node to see only its relations. Since growers take part in almost everything, the **Hide the omnipresent node** box reveals the inner structure.

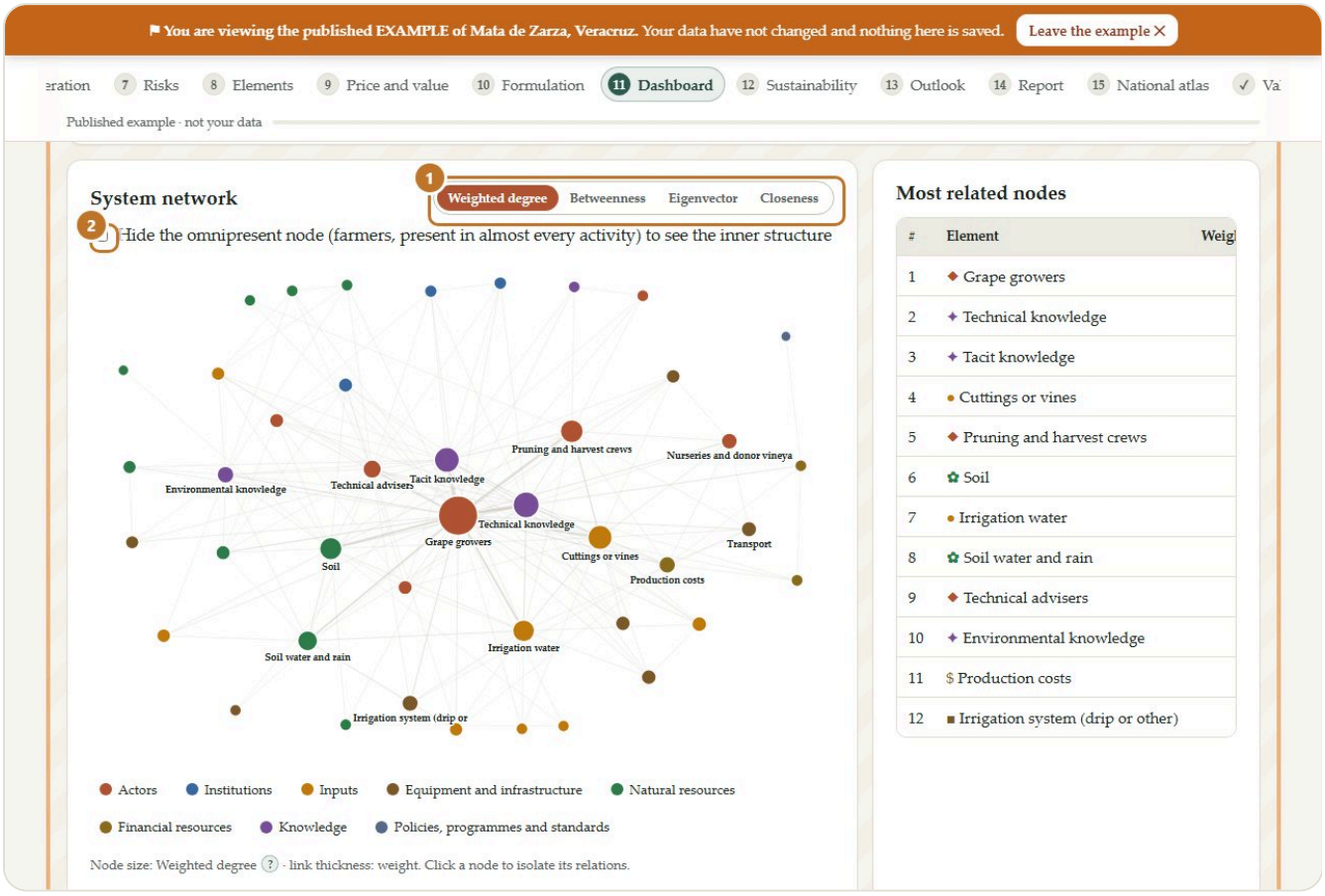


FIGURE 11.2 **System network.**

1. Measure that sets node size.
2. Hide growers to see the inner structure.

IN MATA DE ZARZA

With growers, the network has 42 elements and 175 links, and its groups barely show (Q = 0.06). Hiding them leaves 41 elements and 136 links in **four communities** (Q = 0.19): **soil, water and nutrition** (mulch, compost, bio-inputs, fertilisers, irrigation, soil, moisture, cover, beneficial fauna); **knowledge and institutions** (students, advisers, researchers, universities, INIFAP, associations, programmes, credit, technical knowledge); **young-vine care and local knowledge** (labourers, family, guards, stakes, trellis, weeds, leaf-cutter ants, tacit and cultural knowledge); and **planting material** (nurseries and donor vineyards, cuttings, river sand, rooting chamber, transport, costs). Technical knowledge, tacit knowledge, soil and cuttings are the biggest bridges.

11.3 One element's correlations

Choose an element in the list to see which ones appear together with it in the same activities (green bars) and which hardly ever coincide (red bars).

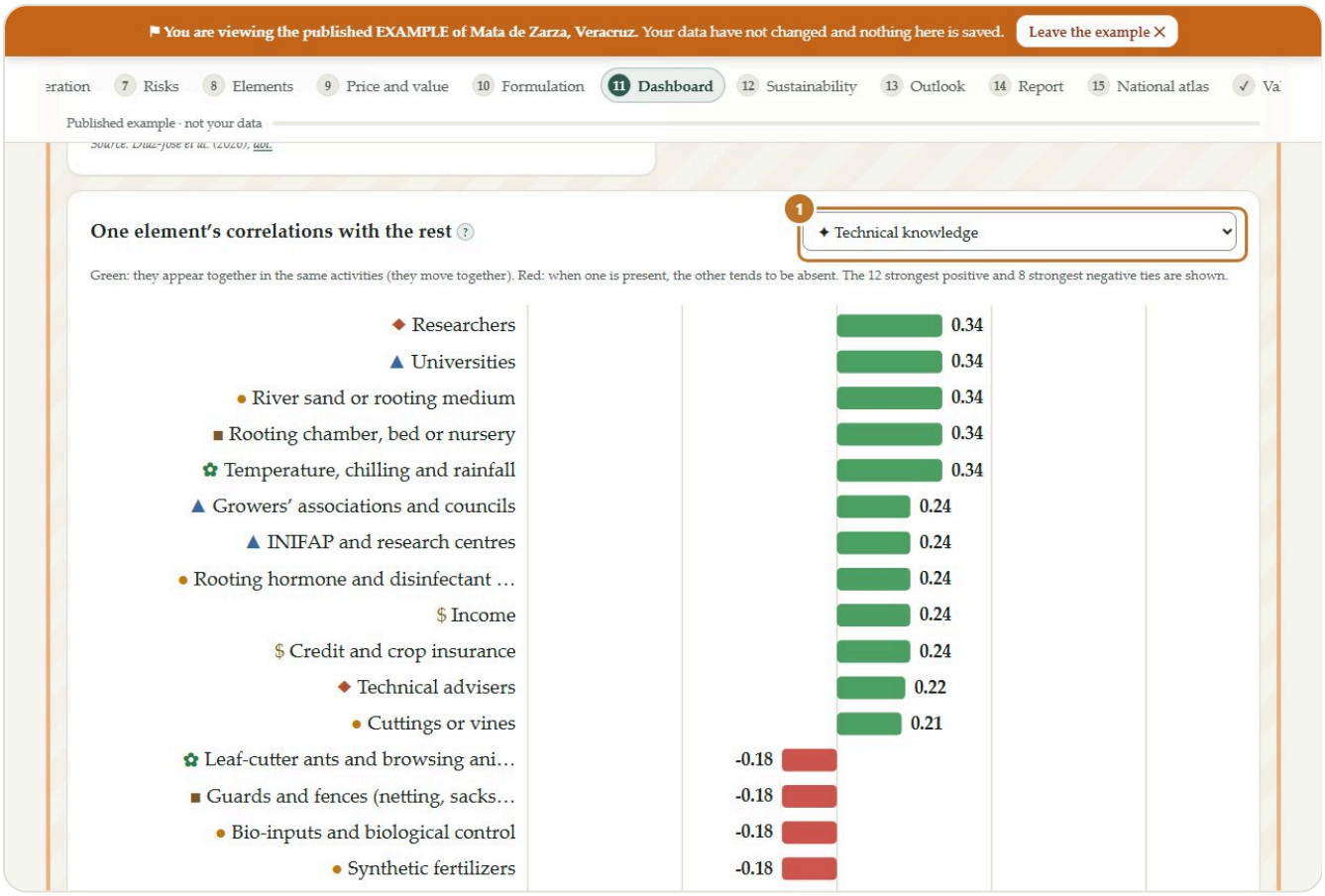


FIGURE 11.3 **One element's correlations.**

1 Element to analyse.

11.4 The cladogram: who is close to whom?

The cladogram groups elements by closeness in the network: each first joins its **nearest neighbour**. Read the tree left to right: short branches are tightly linked elements, best looked after together. Choose to see actors only, actors and institutions, or all elements. Below, the table gives each one's nearest neighbour; ↔ marks mutual neighbours.

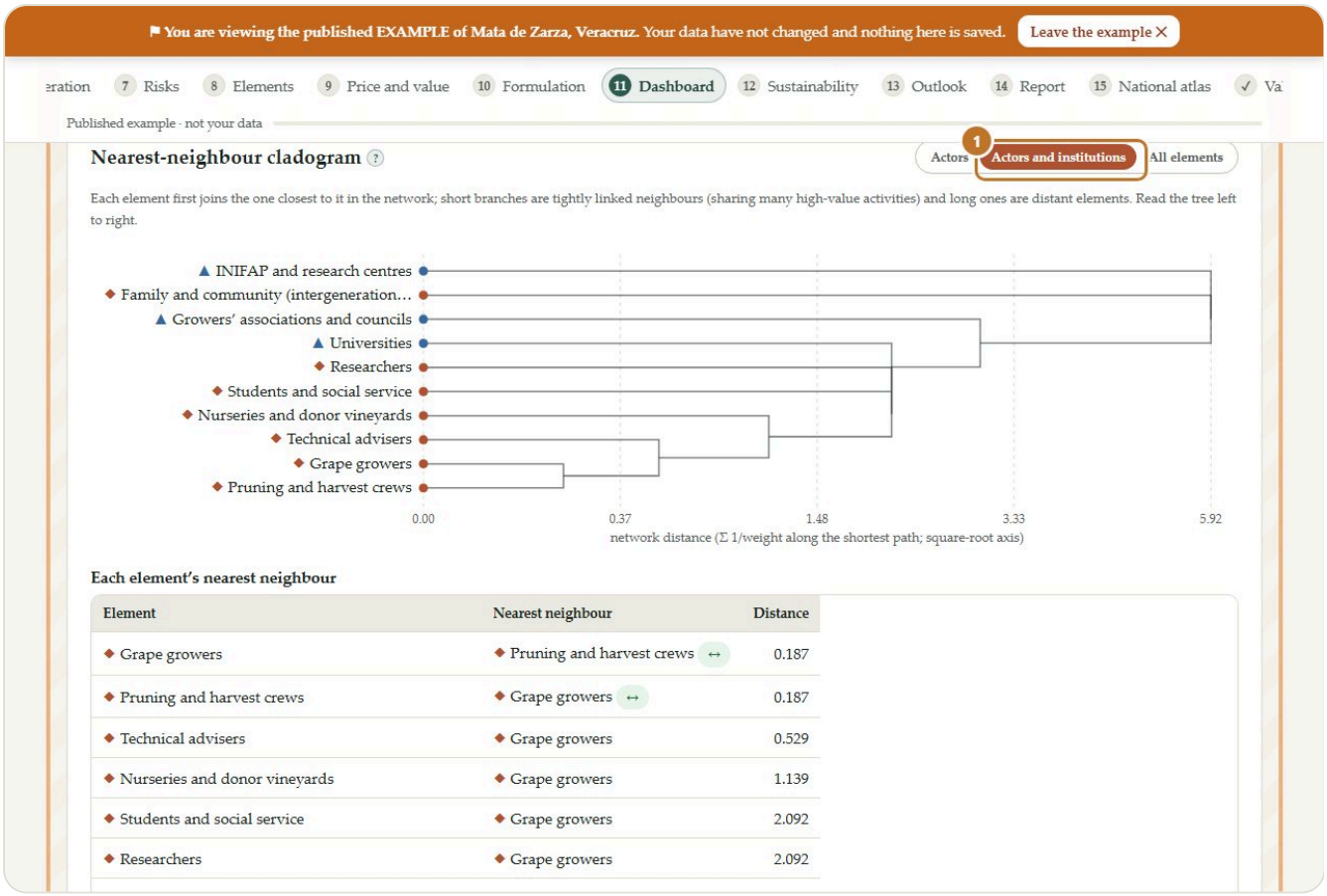


FIGURE 11.4 **Mata de Zarza actors and institutions cladogram.** Growers and labourers are mutual neighbours (0.19).

1 Which elements to group.

> **!** **DISTANT JOINS**
>
> An element taking part in few activities joins late and at a great distance. If an actor important to your vineyard shows up far away, check that it is chosen in every activity it takes part in.

11.5 The circular calendar

The year is a circle; each ring is an activity type and each arc a step with its dates. The needle marks today and the centre announces the next activity. Tap an arc to open its step in Block 5; tap the legend to filter.

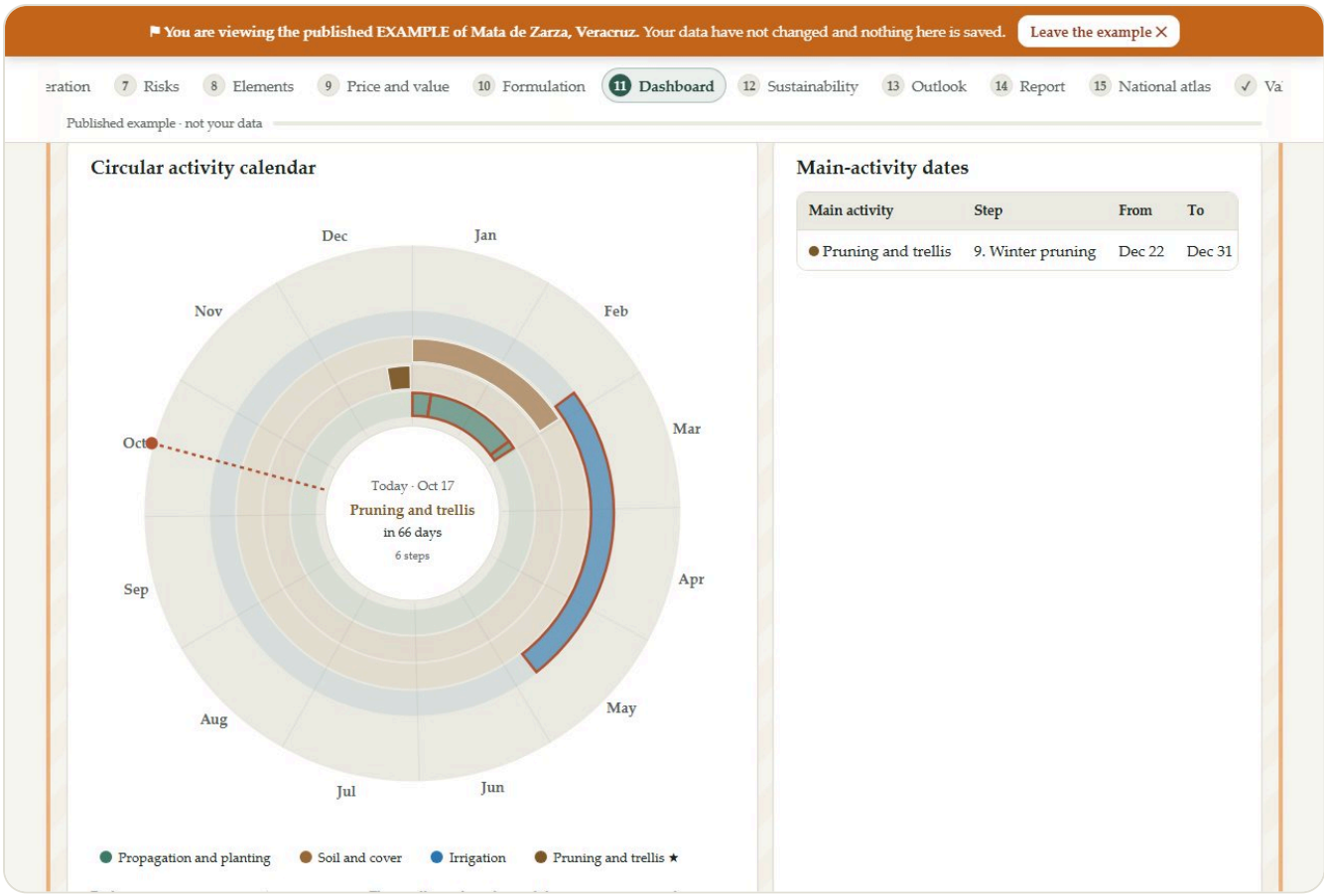


FIGURE 11.5 **Mata de Zarza circular calendar.** From taking cuttings in January to late-December pruning.

BLOCK 12 · SUSTAINABILITY

12 Soil, water, SDGs and footprint

Every activity leaves a mark. Here you see which add, which subtract, Sarandón's index, the 17 SDGs and the carbon footprint.

12.1 Sustainability index and Sarandón

The **sustainability index (IS)** goes from 0 to 1: 0.5 is neutral, above it the activity adds and below it subtracts. **Sarandón's general sustainability index (ISGen)** averages three dimensions on a 0–4 scale: economic (IK), ecological (IE) and socio-cultural (ISC). The system is sustainable if ISGen is above 2 and no dimension falls below 2.

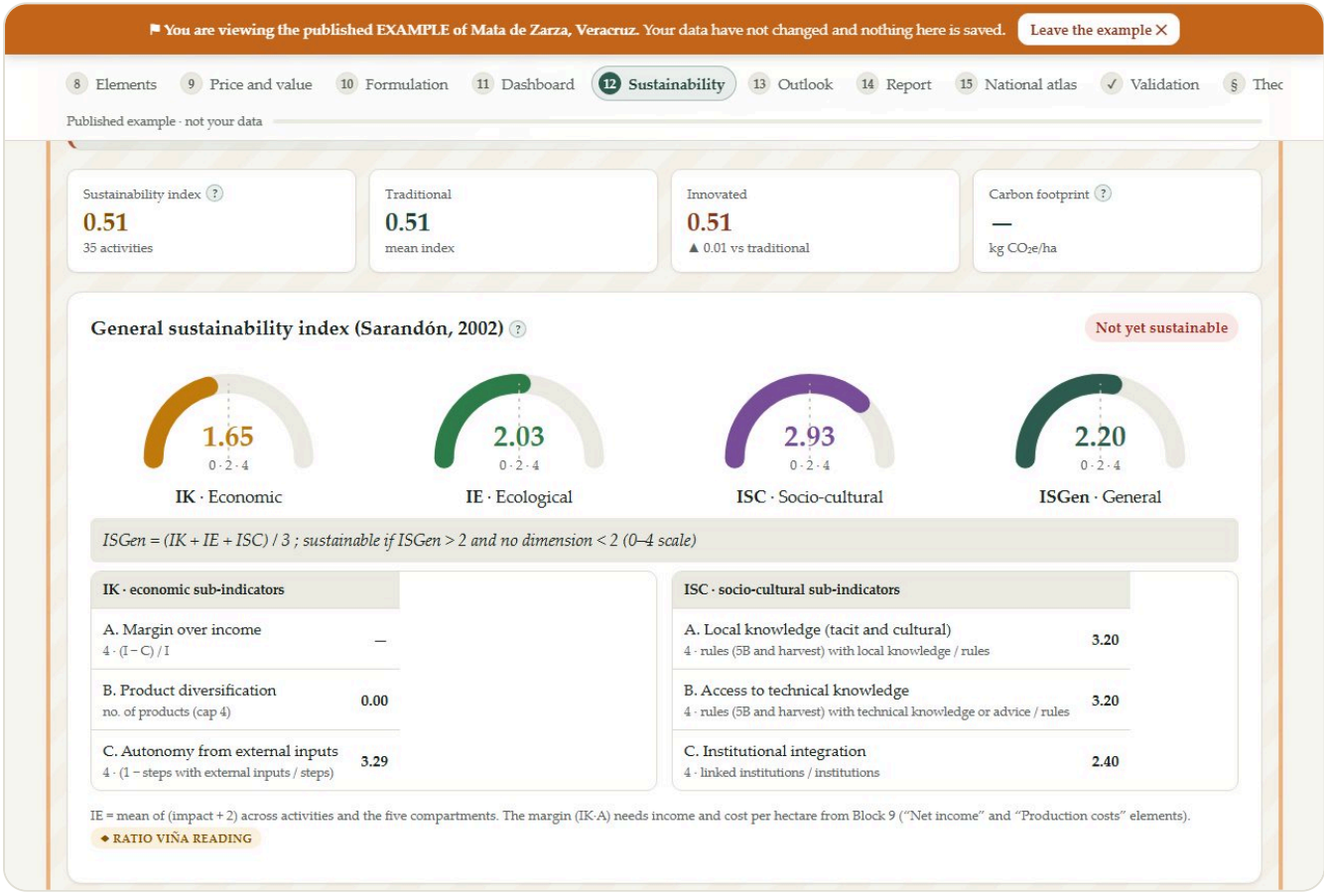


FIGURE 12.1 **Mata de Zarza general sustainability index.** IK 1.65, IE 2.03, ISC 2.93 and ISGen 2.20.

12.2 The 17 SDGs

The wheel and cards show how far each Sustainable Development Goal is met with evidence from your system. “No data” is not failure: it means your data says nothing about that goal.

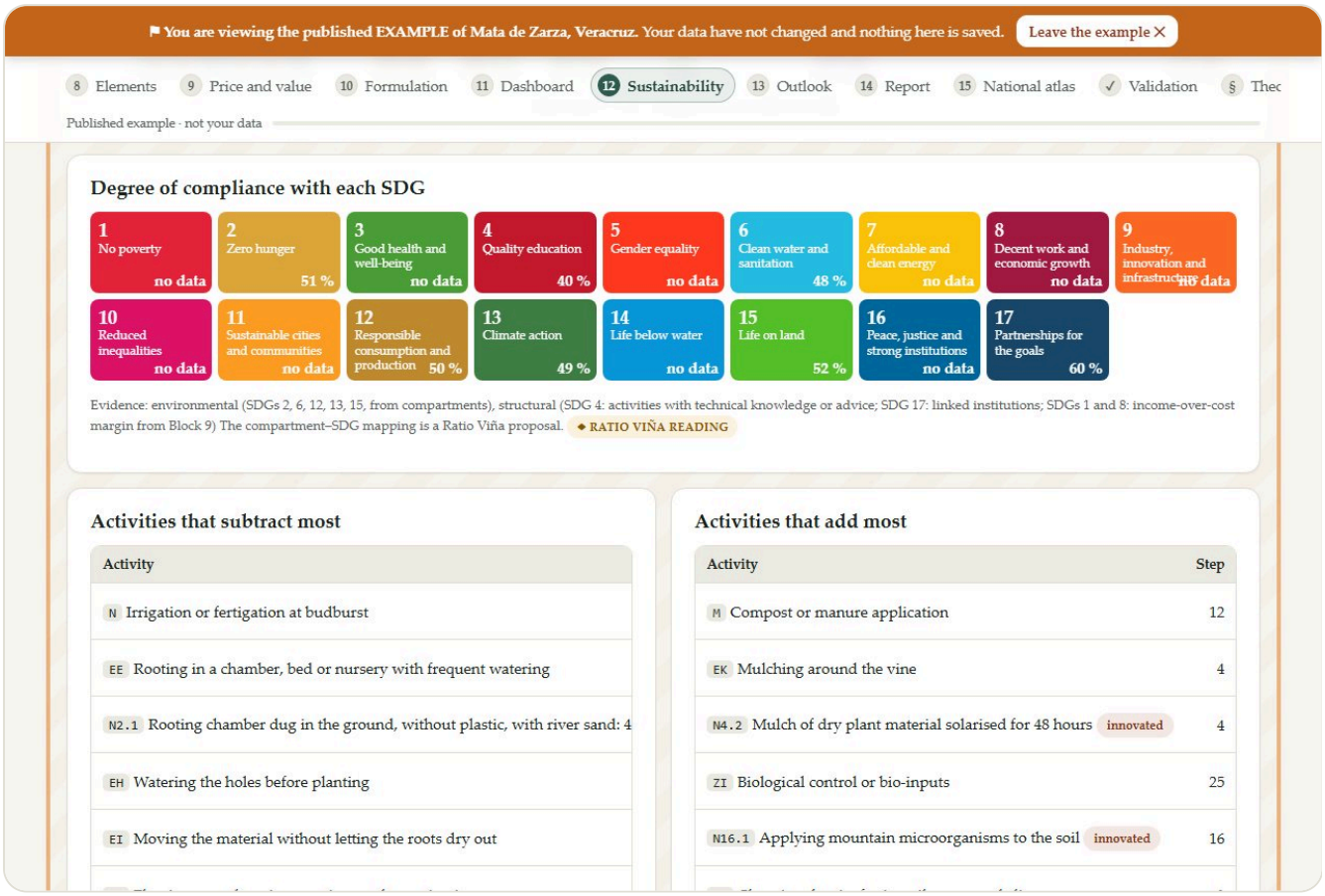


FIGURE 12.2 **SDGs in Mata de Zarza.** Seven of the 17 have evidence.

12.3 Carbon footprint

Enter litres of diesel, kilos of nitrogen and urea per hectare and the yield: the app computes kilos of CO_2 equivalent per hectare and per tonne with IPCC factors. If you leave the fields empty, the footprint shows as “—”.

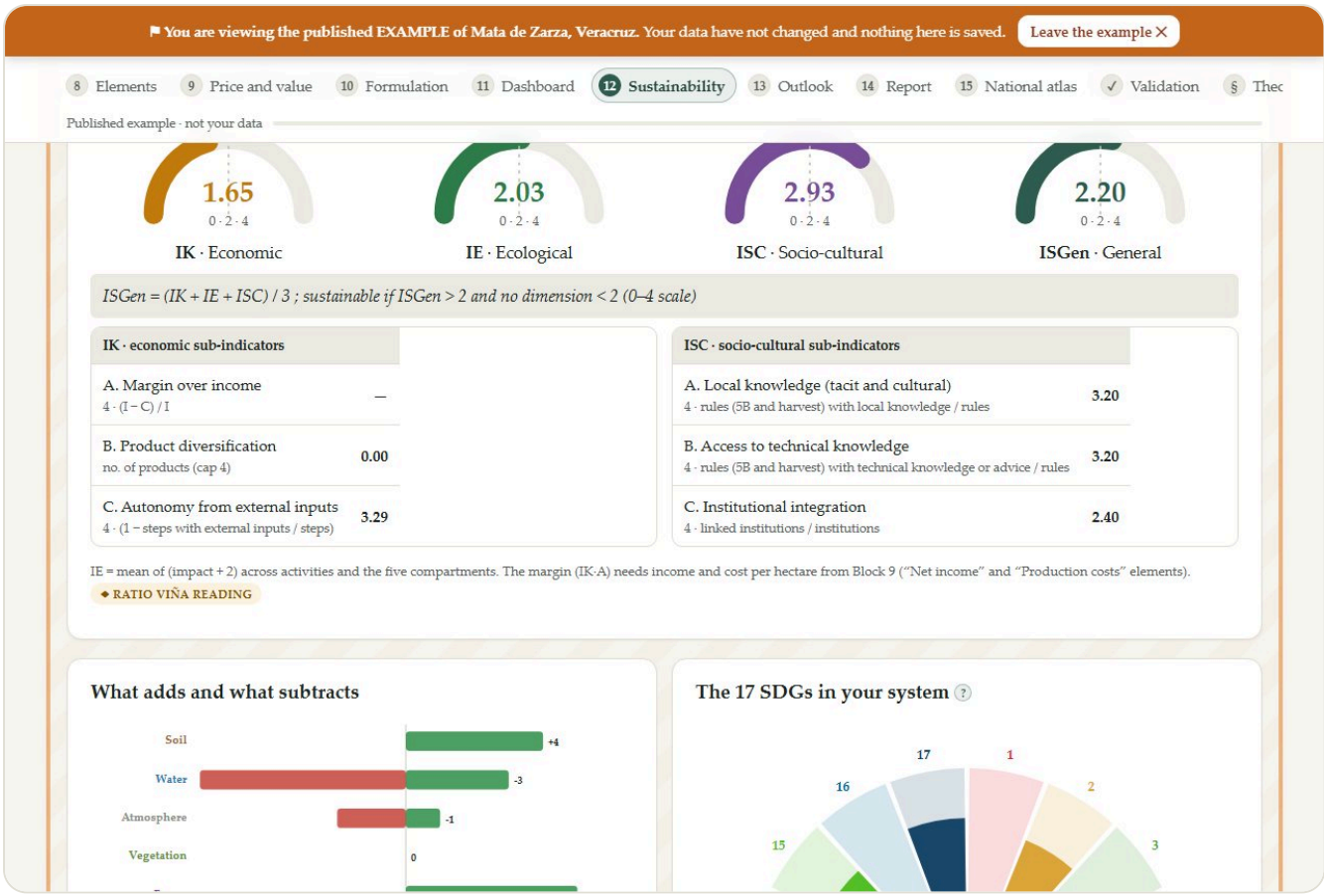


FIGURE 12.3 **What adds and what subtracts** by compartment.

◆ **IN MATA DE ZARZA**

The mean IS of the 35 activities is **0.51**, almost neutral. Those adding most are compost or manure and mulching; those subtracting most are the irrigation ones, because of the water drawn, though drip partly offsets it. The ecological (2.03) and socio-cultural (2.93) dimensions exceed 2, but the economic one stays at **1.65**: the plot does not sell grapes yet, has a single product and relies on cuttings, transport and inputs from outside. So the ISGen of 2.20 does not yet meet the sustainability condition. Once the first harvest's income is entered, the economic dimension is recomputed. The guide publishes no litres of diesel or kilos of nitrogen per hectare, so the footprint awaits those data.

BLOCK 13 · OUTLOOK

13 Three futures at 2, 5 and 10 years

They are not forecasts: they are scenarios to discuss with the family, advisers and community what to do today.

13.1 Desirable, trend and undesired

Each scenario combines product price, input cost, warming, frequency of extreme years, available labour and adoption of your innovative activities. The app plots ten indicators marked at years 2, 5 and 10: yield, price, cost, gross income ÷ cost, net income, labour, adoption, sustainability, regeneration and risk. Below, the **potentials** and **undesired prospects** are written from your results.

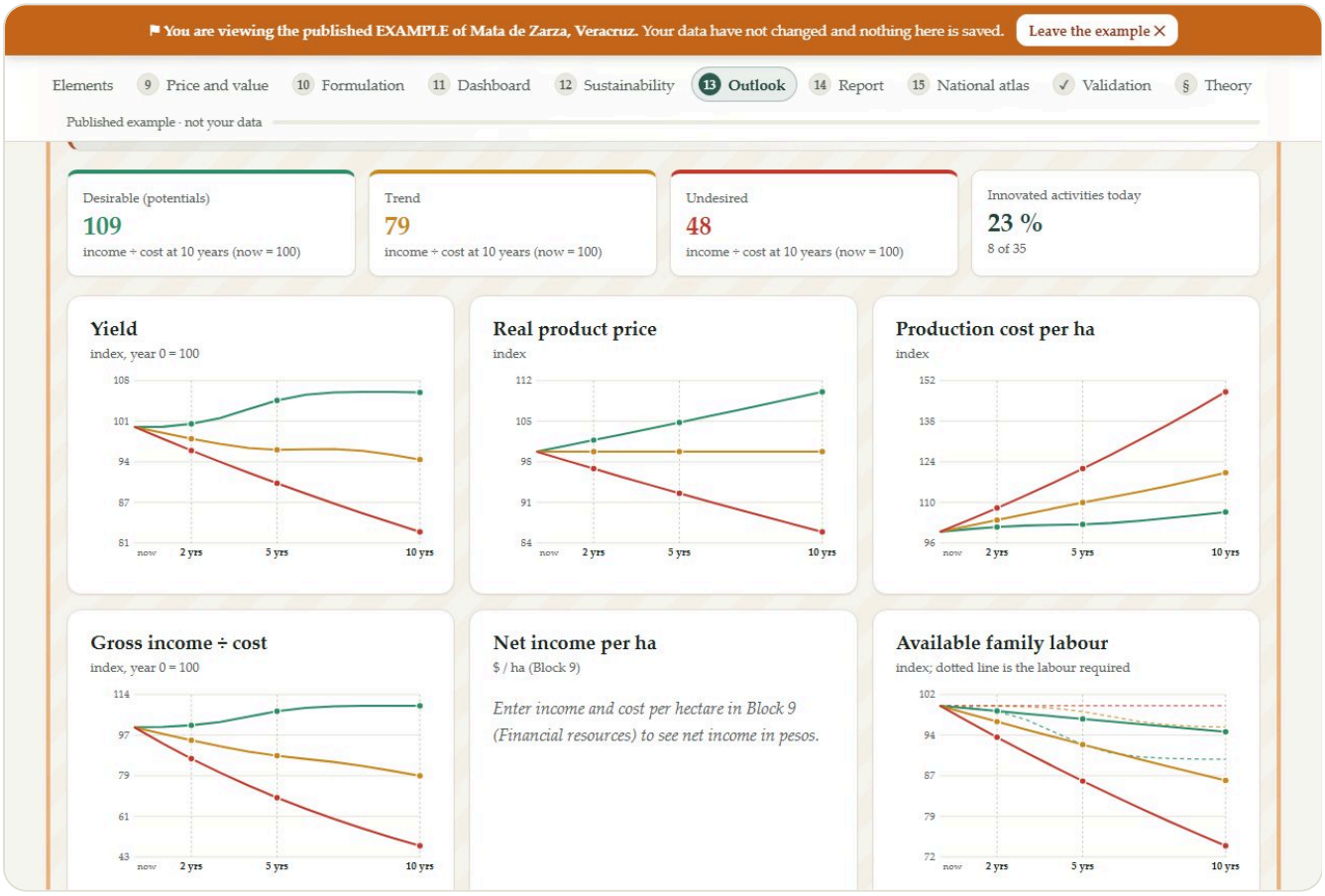


FIGURE 13.1 **Block 13.** The case's indicators at 10 years in the three scenarios.

◆ **IN MATA DE ZARZA**

With the default assumptions and today = 100, at 10 years the gross income ÷ cost ratio is **109** (desirable), **79** (trend) and **48** (undesired), and yield 106, 95 and 83. As the case has 8 innovative activities (23 % of those practised), adoption matters: it reaches 90 % in the desirable scenario, 50 % in the trend one and stays at 23 % in the undesired one, and with it the regeneration index goes from 50 to 67, 57 and 50. These are scenarios with general assumptions, not a projection from the guide; for the Mata de Zarza vineyard, price and cost are best adjusted with first-harvest data.

13.2 The assumptions

The assumptions table is editable. What has a published basis carries its source (warming, IPCC 2018; yield loss per degree, Zhao and colleagues 2017; adoption curve, Rogers 2003); the rest is flagged ◆ as an assumption. Check prices and costs against SIAP and FIRA's reference costs for grapes.

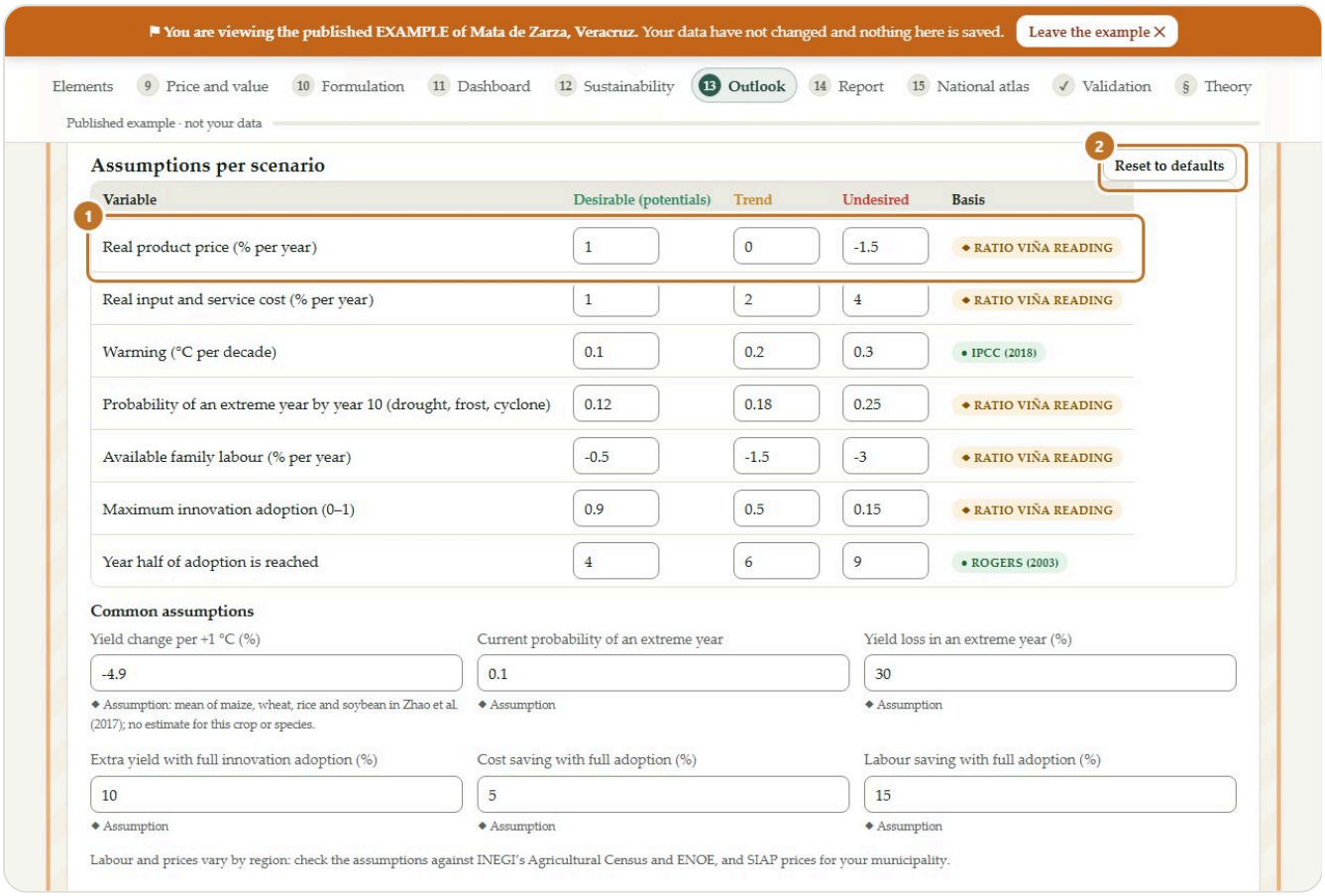


FIGURE 13.2 **Assumptions for each scenario.**

1. Each row is an assumption; each column, a scenario.
2. **Reset to defaults**.

BLOCK 14 · REPORT

Your whole vineyard in one document

14

Written summary, territory, climate, history, calendar, key elements, cladogram, regeneration, risks, outlook and SDGs, ready to print.

14.1 Print, export and save

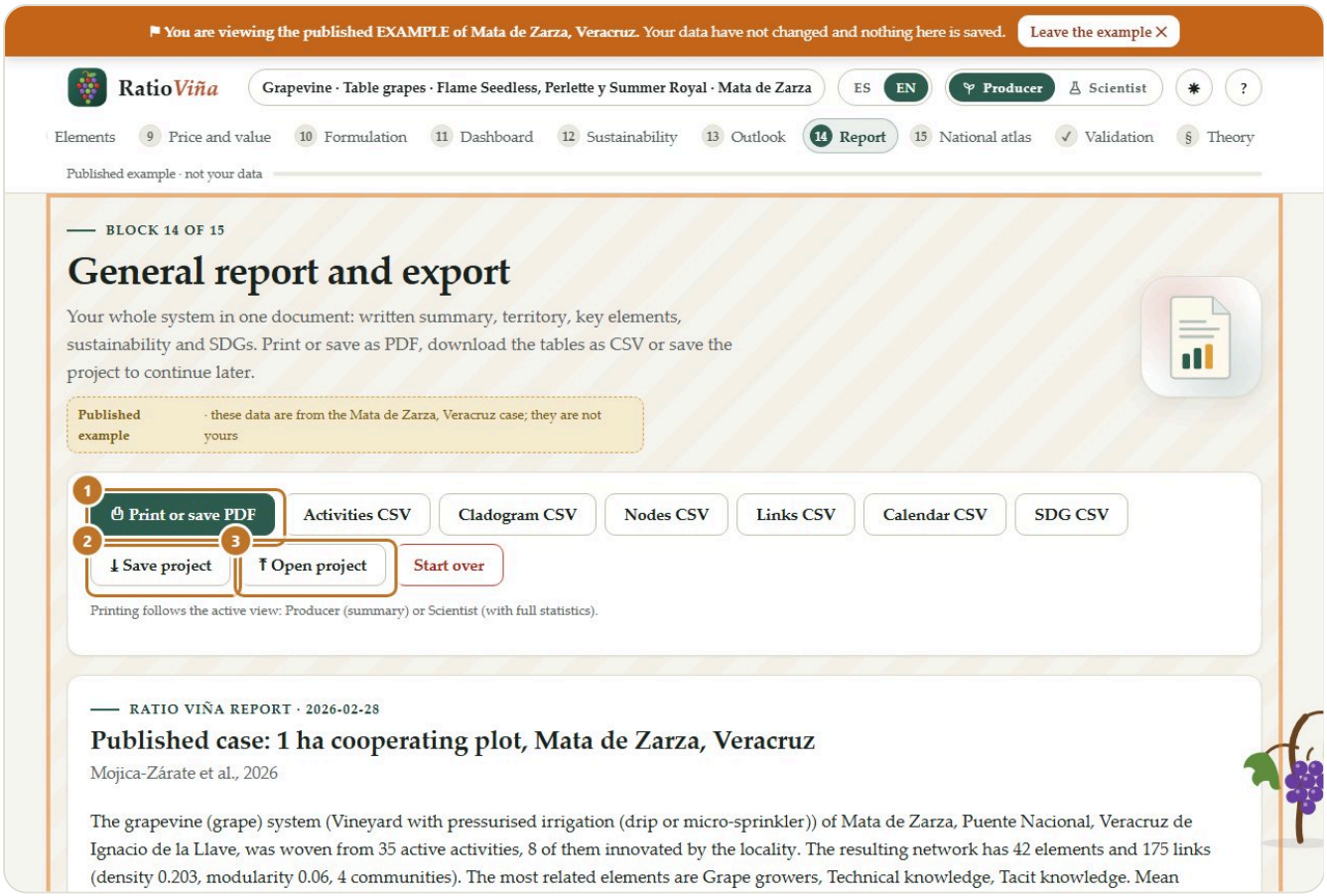


FIGURE 14.1 **Block 14.**

1. **Print or save PDF**: in the print dialog choose “Save as PDF”. Printing follows the active view.
2. **Save project**: a `.vina.json` file with everything you entered.
3. **Open project**: loads a saved file. Version 0.1.0 projects open without losing answers: their steps shift automatically after the new establishment stage.

The CSV buttons download the activities, nodes, links, calendar, SDG and cladogram tables for use in a spreadsheet. **Start over** erases the project from this browser: save it first.

> ✓ **PRINTING IS ALREADY LOOKED AFTER**
>
> When printing, the app always uses the light theme, expands the theory, drops animations and fits tables to the page width, even in night mode.

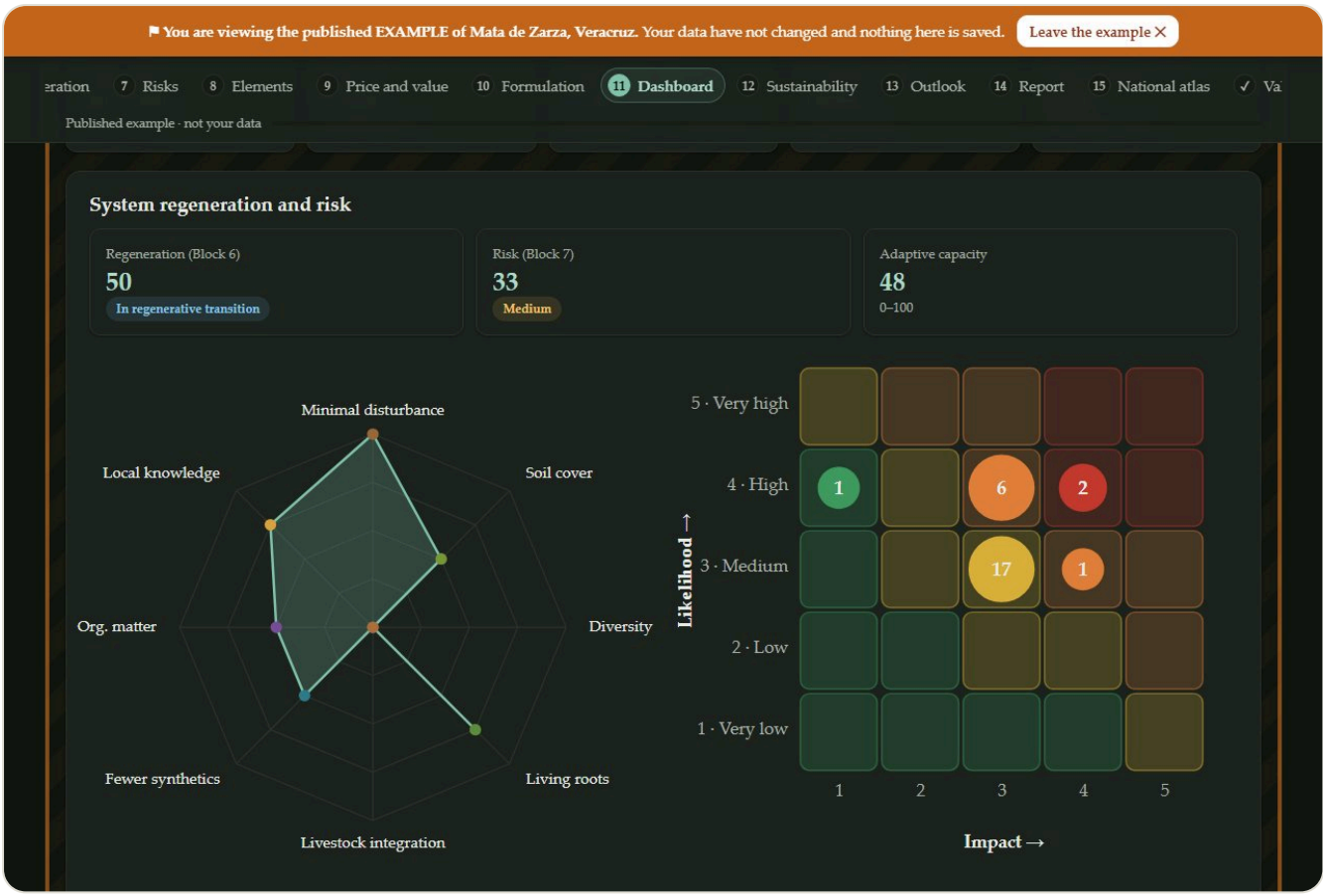


FIGURE 14.2 **Night mode.** On screen you can work at night; the printed report comes out light.

BLOCK 15 · NATIONAL ATLAS

15 Many vineyards, one atlas

The atlas combines the files of the documented vineyards to read local grapevine knowledge at national scale, without a server.

15.1 How the national study is put together

1. **Each surveyor** fills in one vineyard, ideally with those who work it, and saves it in Block 14 with ⤓ Save project: the result is a `.vina.json` file named after the vineyard.
2. **The coordinator** gathers the files (USB stick, e-mail, shared folder) and in Block 15 taps + Add files; many can be chosen at once.
3. With ⤓ Export atlas everything is saved in a single `.atlas.json` file that can be shared or extended; the vineyards and practices CSVs open in a spreadsheet.

> ? **WHAT ABOUT PRIVACY?**
>
> No data goes to a server: the atlas lives in the coordinator's browser and in the files you choose to share. Ask farmers' permission before writing their names; the atlas only needs the locality and its answers.

15.2 What the atlas shows

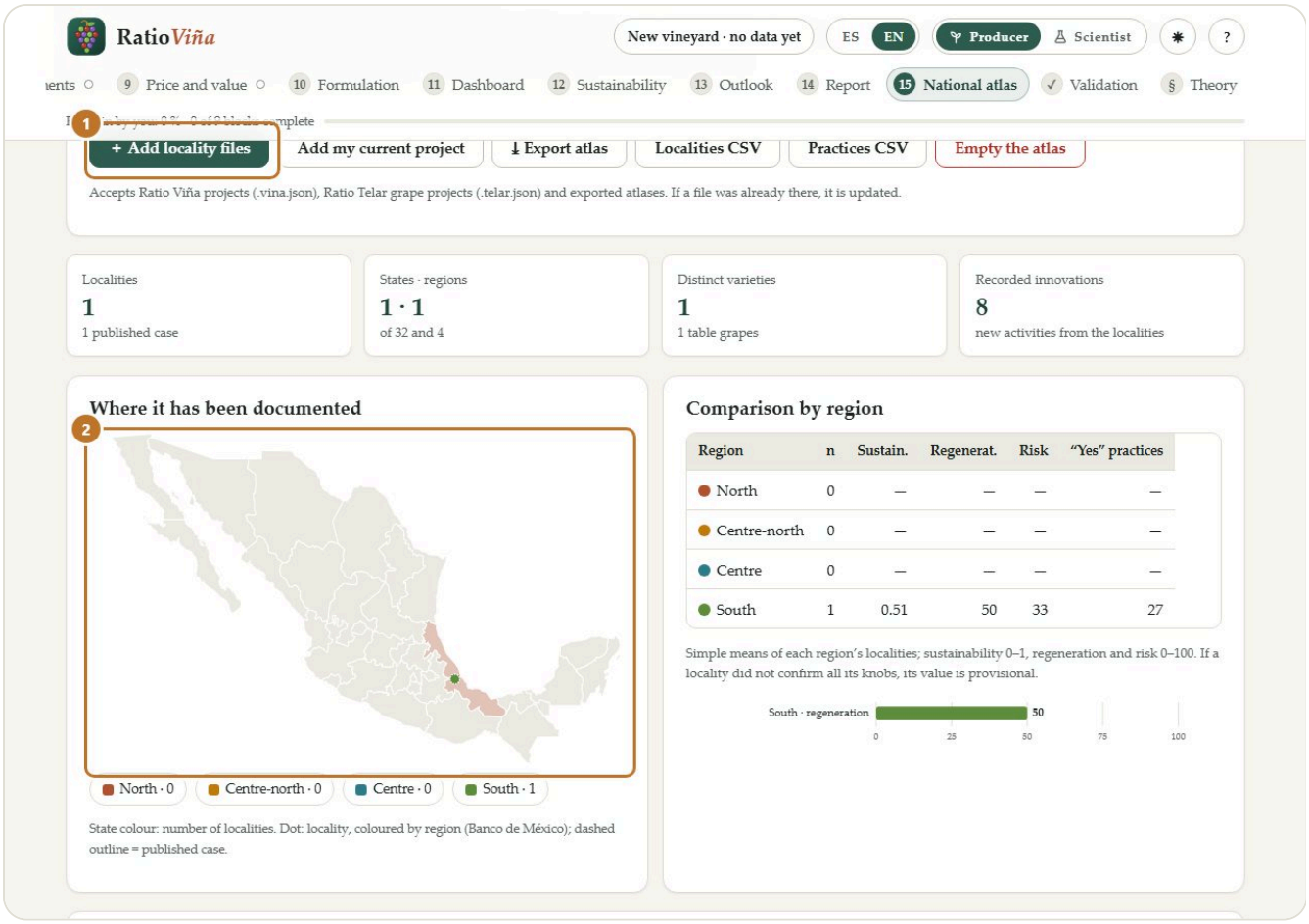
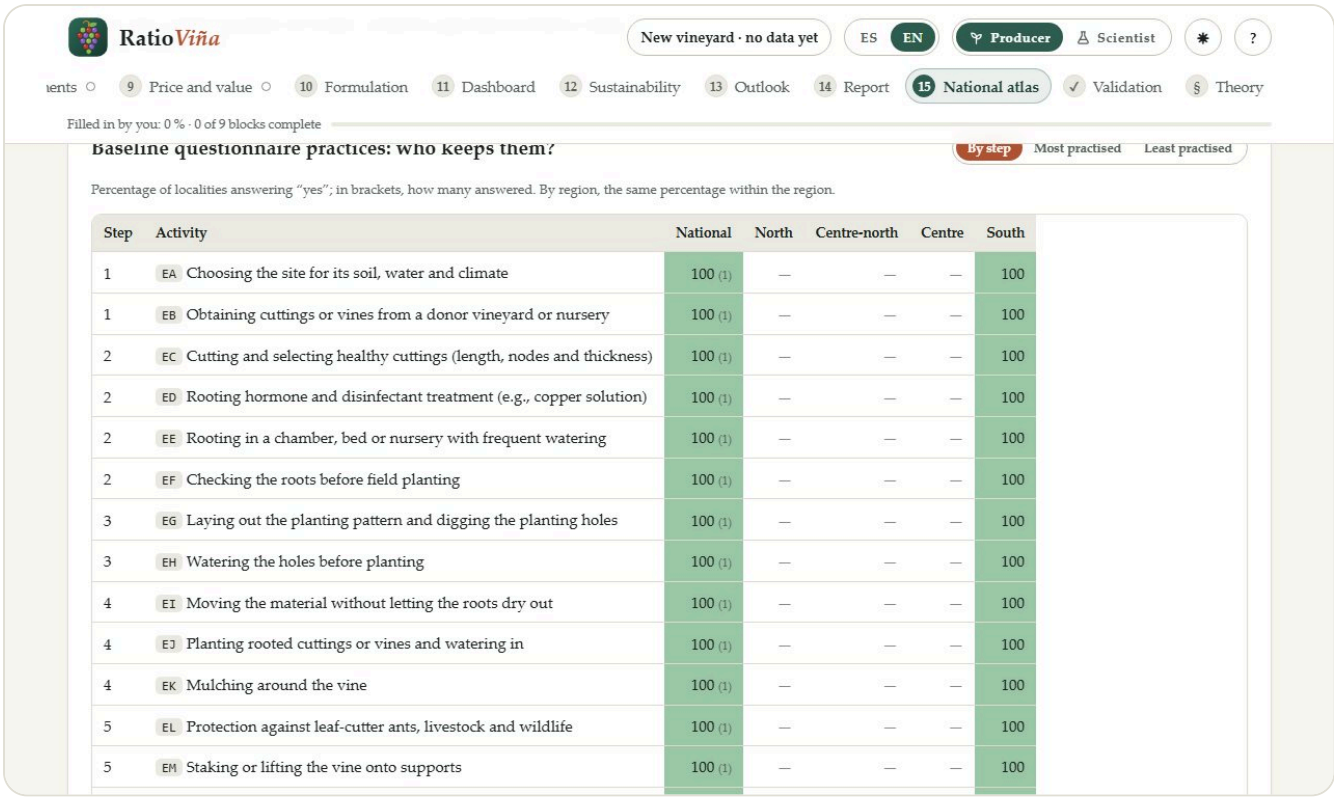


FIGURE 15.1 **National atlas** with the Mata de Zarza case.

1. Buttons to add files, add your project and export.
2. Map: the state colour shows how many vineyards there are; each dot is a vineyard.

- **Comparison by region** (North, North-central, Central and South), by **use** (table, wine, raisins, juice) and by **Winkler region**: number of vineyards and means of sustainability, regeneration, risk and practices answered "yes".
- **Baseline questionnaire practices**: for each activity, the share of vineyards that practise it, nationally and by region.
- **Varieties**, planting dates and number of prunings, vineyards' **innovations** and the **national network** of grapevine knowledge.



Step	Activity	National	North	Centre-north	Centre	South
1	EA Choosing the site for its soil, water and climate	100 (1)	—	—	—	100
1	EB Obtaining cuttings or vines from a donor vineyard or nursery	100 (1)	—	—	—	100
2	EC Cutting and selecting healthy cuttings (length, nodes and thickness)	100 (1)	—	—	—	100
2	ED Rooting hormone and disinfectant treatment (e.g., copper solution)	100 (1)	—	—	—	100
2	EE Rooting in a chamber, bed or nursery with frequent watering	100 (1)	—	—	—	100
2	EF Checking the roots before field planting	100 (1)	—	—	—	100
3	EG Laying out the planting pattern and digging the planting holes	100 (1)	—	—	—	100
3	EH Watering the holes before planting	100 (1)	—	—	—	100
4	EI Moving the material without letting the roots dry out	100 (1)	—	—	—	100
4	EJ Planting rooted cuttings or vines and watering in	100 (1)	—	—	—	100
4	EK Mulching around the vine	100 (1)	—	—	—	100
5	EL Protection against leaf-cutter ants, livestock and wildlife	100 (1)	—	—	—	100
5	EM Staking or lifting the vine onto supports	100 (1)	—	—	—	100

FIGURE 15.2 **Who keeps each practice?** Share of vineyards answering "yes", nationally and by region.

→ **READING RULE**

A percentage only counts vineyards that **answered** that activity: unanswered items are left out, not taken as "no". That is why each percentage shows, in brackets, how many vineyards answered. With fewer than five, read it as a hint, not a trend.

REFERENCE

Documenting a vineyard for the national study

The full route to document a vineyard so that its file can join the national atlas.

D.1 Eleven steps

1. **Bring people together.** Whoever works the vineyard, whoever will inherit it, the technical adviser. In Mata de Zarza farmers, students and lecturers took part, and what was learnt was systematised as a case study.
2. **Block 1.** Use, variety, rootstock, training, vines per hectare, planting date and material and every pruning.
3. **Block 2.** Locate the locality.
4. **Block 3.** Enter the monthly temperatures of the nearest station and note the source.
5. **Block 4.** Check with people which requirements are met and record those they have learnt.
6. **Block 5.** Answer “yes” or “no” to each activity, set the dates and add the vineyard’s own activities.
7. **Blocks 6 and 7.** Adjust the regeneration and risk knobs to what the vineyard lives; save the baseline.
8. **Block 8.** Check all elements are there; add the missing ones and choose them in their activities.
9. **Block 9.** Enter prices, costs and income per hectare and rate value.
10. **Block 10.** Weave the system and read it in blocks 11–13 with the people who live it.
11. **Block 14.** Print the report and save the project: the `.vina.json` file is the one sent to the atlas coordinator (Block 15).

? **YOUR RECORD IS KNOWLEDGE**

A well-documented vineyard is the first step to publishing its local knowledge Stack, as was done with establishment in Mata de Zarza. Keep the sources of every figure.

D.2 Interview guide, block by block

Open questions for a conversation with those who work the vineyard. The order is not important; what matters is that every answer is written down with who said it and when.

BLOCK	QUESTIONS
1	What do you grow grapes for? Which variety and rootstock, and where did the vines come from? When was it planted and with what material? When and how do you prune?
2	In which locality and municipality is the vineyard? Irrigated or rainfed?
3	Which is the nearest weather station? Are there cold months? When does it rain and when is water short?
4	What does the vine need to do well here? What must never be done? What have you changed and why?
5	What are the year's steps, first to last? What is done in each, with which tools and by whom? In which months?
6	How many times is the soil worked? Is it covered? Manure, compost or bio-inputs? Animals? Has the soil been tested?
7	What was the worst thing that happened to the vineyard? How often does it happen? How much was lost? What do you do to prevent it?
8	Who else takes part: family, labourers, nurseries, advisers, buyers, universities, programmes?
9	How much does it cost to establish and maintain a hectare and what are you paid for the grapes? What is this element worth to you, beyond money?
13	Who will look after the vineyard in ten years? What would you like to change and what not?

✓ **CONSENT**

Explain what the data will be used for and ask permission before writing down names. In the report the locality and the number of informants are enough.

D.3 Field sheet 1: steps and activities

Print this page and fill it in by hand in the vineyard; then transfer it to Block 5.

Vineyard: _____ Locality: _____ Date: _____

#	STEP	ACTIVITY	WHO · WITH WHAT	MONTH	T / I
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					

T = traditional, I = innovated.

D.4 Field sheet 2: climate, planting and prunings

Monthly temperatures in °C (normals of the nearest station). Source: _____

	E/J	F	M	A	M	J	J	A	S	O	N	D
Maximum												
Minimum												
Mean												

Planting date: ____ / ____ / _____ Material: _____ Vines per ha: _____

PRUNING DATE	TYPE	NOTE (BUDS, CREW, OBSERVATIONS)

D.5 Field sheet 3: regeneration

Mark a number from 0 to 4 on each row with the people and write why; then transfer it to the Block 6 knobs.

INDICATOR	0	1	2	3	4	WHY?
Minimal tillage	○	○	○	○	○	
Soil cover	○	○	○	○	○	
Diversity	○	○	○	○	○	
Living roots	○	○	○	○	○	
Livestock integration	○	○	○	○	○	
Fewer synthetic inputs	○	○	○	○	○	
Organic matter	○	○	○	○	○	
Local knowledge	○	○	○	○	○	

Sum: _____ ÷ 32 = IR _____ Organic matter from soil test: _____ % Year: _____

D.6 Field sheet 4: risks

For each hazard: likelihood (1 rarely … 5 almost every year), impact (1 almost nothing … 5 everything lost) and control (0 none … 3 well controlled). Cross out those that do not apply.

HAZARD	LIK.	IMPACT	CONTROL	WHAT WE DO
Drought or late rains				
Frost				
Hail				
Heavy rain or flooding				
Heatwave				
Cyclones and hurricanes				
Lack of winter chill				
Pests				
Fungal diseases				
Pierce's disease				
Leaf-cutter ants				
Animal damage				
Diseased or poorly rooted material				
Herbicide drift				
Resistant or invasive weeds				
Fall in sale price				
Rise in input and service costs				
New crop without a market				
Middlemen or no buyer				
Labour shortage and no succession				
Abandoning the crop				
Loss of local knowledge				
Insecurity or theft				
Withdrawal of programmes and support				
Land tenure conflicts				
Erosion and fertility loss				
Water scarcity				

HAZARD	LIK.	IMPACT	CONTROL	WHAT WE DO
Clay or poorly drained soils				

D.7 Field sheet 5: costs, price and value

Per hectare and per cycle. Then transfer it to Block 9.

ITEM	QUANTITY	UNIT PRICE	$
Cuttings or vines and their transport			
Posts, wires and irrigation system			
Fertilisers, manures and bio-inputs			
Fungicides, insecticides and herbicides			
Labour days (family and paid)			
Water, fuel, packing, freight, other			
Total cost			
Grapes sold (kg or boxes)			
Wine, raisins, juice, cuttings sold or other			
Total income			

REFERENCE

E Troubleshooting

If something does not look as you expect, look here.

I SEE THIS	DO THIS
The Dashboard says “You have not woven the system yet”.	Go to Block 10 and tap **Weave the system**.
The climate indices say “—”.	Temperatures are missing: Winkler needs April–October means; Huglin, April–September means and maxima; the cool-night index, the September minimum.
The vineyard age says “—”.	Move the planting-date knob and type the year, or type the age in years.
An element I added is not in the network.	Choose it in at least one activity.
Net income says “no data”.	Enter income and cost per hectare in Block 9 › Financial resources.
The carbon footprint says “—”.	Enter diesel, nitrogen, urea and yield in Block 12.
I opened a version 0.1.0 project and the steps changed number.	That is expected: the establishment stage now takes steps 1–6 and your answers moved to steps 7–37.
I changed computer and my data is gone.	Data lives in each browser: use Report › Save project and Open project.
I do not see a new version’s changes.	Reload with Ctrl + F5.

REFERENCE

F Glossary

The app's and the vineyard's words, in plain language.

Activity *Actividad*	What is done within a management step (for example, "watering the holes before planting"). Each activity links the elements taking part in it.
Innovative activity *Actividad innovadora*	A new practice the vineyard added to the baseline questionnaire; in the app it is added with "+ Innovative activity".
Mulch *Acolchado*	A layer of dry plant material around the vine that keeps moisture in and protects the soil.
Rooted cutting *Barbado*	A cutting that has already rooted and is planted in its final place.
BBCH *BBCH*	Scale numbering the vine's development stages, from winter bud to leaf fall (Lorenz et al., 1995).
Rooting chamber *Cámara de enraizamiento*	A pit in the ground, filled with moist river sand, where cuttings form roots for about 45 days before planting (Mojica-Zárate et al., 2026).
Adaptive capacity *Capacidad adaptativa*	How well the vineyard can absorb a shock and reorganise. Ratio Viña measures it with regeneration, diversification, technical knowledge and institutions.
Planting hole *Cepa de plantación*	Hole where the cutting is planted; in Mata de Zarza, 20 cm wide and 70 cm deep.
Cladogram *Cladograma*	Tree that groups elements by how close they are in the network: short branches are tightly linked neighbours.
Community *Comunidad*	Group of elements more related to each other than to the rest of the network.
Training system *Conducción*	The shape given to the vine on its structure: vertical trellis, cordon, pergola, lyre or bush vine.

Element *Elemento*	Everything taking part in the system: actors, institutions, inputs, equipment, natural and financial resources, knowledge, policies, disturbances and products.
Veraison *Envero*	The moment the berry changes colour and starts ripening.
Scenario *Escenario*	A possible future built on explicit assumptions. It is not a forecast.
Cutting *Esqueje*	A piece of cane from pruning, about 70 cm long with at least six nodes, rooted to obtain a new vine.
Cool-night index *Frescor nocturno*	Mean minimum temperature of the ripening month; it says how cool the nights are (Tonietto and Carbonneau, 2004).
Winkler degree-days *Grados día de Winkler*	Sum of heat useful to the vine from April to October, above 10 °C (Amerine and Winkler, 1944).
Carbon footprint *Huella de carbono*	Greenhouse gases emitted per hectare (diesel, nitrogen and urea), in kg of CO_2 equivalent.
Huglin index *Índice de Huglin*	Sum of April–September heat that weighs sunny hours more (Huglin, 1978).
Regeneration index *Índice de regeneración*	Average of the eight regeneration knobs, from 0 to 100. Tells whether the vineyard gives back more than it takes.
Risk index *Índice de riesgo*	Average risk of all applicable hazards, from 0 to 100.
Sustainability index (IS) *Índice de sostenibilidad (IS)*	How much each activity adds to or subtracts from soil, water, atmosphere, vegetation and fauna, from 0 to 1 (0.5 is neutral).
Betweenness *Intermediación*	How many short network paths pass through an element: it measures who acts as a bridge.
ISGen *ISGen*	Sarandón's general sustainability index: average of the economic, ecological and socio-cultural dimensions, from 0 to 4.
Ratio Viña reading *Lectura de Ratio Viña*	A value or grouping proposed by the app, not published in the source. It is always flagged and can be changed.

Baseline *Línea base*	A snapshot of the vineyard on a date, to compare later and see whether it improves.
Risk matrix *Matriz de riesgos*	A 5 × 5 table crossing how likely a risk is with how much damage it would do.
Mountain microorganisms *Microorganismos de montaña*	A bio-input made from microorganisms collected from forest soil and multiplied on the farm, applied to the soil.
Modularity (Q) *Modularidad (Q)*	How marked the network's groups (communities) are. Above 0.3, groups are clear.
SDG *ODS*	Sustainable Development Goals of the United Nations 2030 Agenda.
Management step *Paso de manejo*	A stage of the vineyard cycle (for example, "propagating cuttings"). The baseline questionnaire has 37.
Knob *Perilla*	A slider you move like a volume control, from a low value to a high one.
Own-rooted vine *Pie franco*	A vine growing on its own roots, without a rootstock.
Training pruning *Poda de formación*	Pruning in the first years that shapes the vine; in Mata de Zarza, the first-year one is done from the base of the stem.
Rootstock *Portainjerto*	The plant that provides the roots when the variety is grafted onto another.
Foresight *Prospectiva*	Discipline that explores possible futures to decide better today.
Reanalysis (ERA5) *Reanálisis (ERA5)*	A reconstruction of past climate with a model and all available observations; it gives data for any point, even without a station (Hersbach et al., 2020).
Weighted network *Red ponderada*	Network in which each link has a thickness: thicker if the elements share many valuable activities.
Regeneration *Regeneración*	The vineyard improving, each cycle, the soil, water and biodiversity it depends on.

Complex adaptive system *Sistema complejo adaptativo*	A set of interacting agents and resources that learn and adapt; behaviours nobody designed emerge from their relationships.
Sustainability *Sostenibilidad*	Producing today without compromising the capacity to produce tomorrow, environmentally, economically and socially.
Local knowledge stack *Stack de conocimiento local*	Layered organisation of a locality's knowledge of a crop: rules, steps and activities (Díaz-José et al., 2026).
Value *Valor*	An element's importance to the system in four dimensions: economic, use, cultural and ecological. Not the same as price.

REFERENCE

Ratio Viña on one page

Cut it out and take it to the vineyard.

1

VineUse, variety, rootstock, training, planting and pruning.

2

RegionLocate your vineyard down to the INEGI locality.

3

Viticultural climate Enter monthly temperatures and read the three indices.

4

RequirementsMark whether each requirement is met and add your own.

5

ManagementAnswer yes or no to each activity, add innovations and dates.

6

RegenerationConfirm the eight knobs and save a baseline.

7

RisksUntick what does not apply and move likelihood, impact and control.

8

ElementsReview actors, inputs and resources; add the missing ones.

9

Price and value Enter prices per hectare and rate value from 0 to 5.

10

Formulation⚙ Weave: the app builds the vineyard network.

11

DashboardRead who holds the system together and who is close to whom.

12

SustainabilitySee what adds, what subtracts and Sarandón's index.

13

OutlookCompare the three futures and adjust the assumptions.

14

ReportPrint, export and save your vineyard's file.

15

AtlasCombine vineyard files and read the country.

Scan to open Ratio Viña:
hectortecumshe-ai.github.io/RatioVina

REFERENCE

How to cite, sources and credits

How to cite this guide

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Main sources

- Mojica-Zárate HT, Barrera-Guzmán LÁ, Alejo-Bello S, Partida-Sedas JG, Rojas-Reyes JO, Díaz-José O, Tinoco-Rueda JÁ, Torres-Palacios C. 2026. *Guía práctica para el establecimiento de uva en Mata de Zarza, Veracruz*. Colección Guías Prácticas para Productores, Tomo 5. Universidad Autónoma Chapingo, Texcoco. ISBN 978-607-12-0712-8. https://sidicyt.chapingo.mx/repositorio/2026/guias-2026-005.pdf
- Amerine MA, Winkler AJ. 1944. Composition and quality of musts and wines of California grapes. *Hilgardia* 15(6): 493–675. · Huglin P. 1978. Nouveau mode d'évaluation des possibilités héliothermiques d'un milieu viticole. *Comptes Rendus de l'Académie d'Agriculture de France* 64: 1117–1126. · Tonietto J, Carbonneau A. 2004. A multicriteria climatic classification system for grape-growing regions worldwide. *Agricultural and Forest Meteorology* 124(1–2): 81–97.
- Hersbach H, Bell B, Berrisford P, et al. 2020. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society* 146(730): 1999–2049. https://doi.org/10.1002/qj.3803
- Lorenz DH, Eichhorn KW, Bleiholder H, Klose R, Meier U, Weber E. 1995. Growth stages of the grapevine. *Australian Journal of Grape and Wine Research* 1(2): 100–103. · Winkler AJ, Cook JA, Kliewer WM, Lider LA. 1974. *General viticulture*, 2.ª ed. · Keller M. 2020. *The science of grapevines*, 3.ª ed. · Kok D. 2014. A review on grape growing in tropical regions. *Turkish Journal of Agricultural and Natural Sciences* 1 (Özel Sayı 1).

- Díaz-José O, Mojica-Zárate HT, Díaz-José J, Vilaboa-Arróniz J. 2026. El conocimiento agrícola local: el stack del maíz criollo en Calpan, Puebla. *Agricultura, Sociedad y Desarrollo* 23(3). https://doi.org/10.22231/asyd.v23i3.1822 · INEGI. 2020. Censo de Población y Vivienda; Catálogo Único de Claves Geoestadísticas.
- Sarandón SJ. 2002. El desarrollo y uso de indicadores para evaluar la sustentabilidad de los agroecosistemas. En: Sarandón SJ (ed.), *Agroecología: el camino hacia una agricultura sustentable*. Ediciones Científicas Americanas, La Plata: 393–414.
- Newton P, et al. 2020. What is regenerative agriculture? *Frontiers in Sustainable Food Systems* 4: 577723. · Schreefel L, et al. 2020. Regenerative agriculture – the soil is the base. *Global Food Security* 26: 100404.
- IPCC. 2014. Climate Change 2014: Impacts, Adaptation, and Vulnerability (SPM). · IPCC. 2018. Global Warming of 1.5 °C (SPM). · ISO 31000:2018 · IEC 31010:2019.
- Zhao C, et al. 2017. Temperature increase reduces global yields of major crops in four independent estimates. *PNAS* 114(35): 9326–9331. · Rogers EM. 2003. *Diffusion of innovations*, 5.ª ed.

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Legal page

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